Evidence Reliability and Action Control in Text-Based Retrieval-Augmented Generation: A Structured Scoping Map and Agenda

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Abstract

Text retrieval-augmented generation (RAG) grounds language models in external evidence, but retrieved evidence can be irrelevant, distracting, conflicting, stale, insufficient, or adversarially corrupted. We present a structured scoping map and agenda for robust RAG under unreliable evidence. The paper is not a venue-complete systematic review or field census. Its central contribution is a comparability framework: robust-RAG methods and benchmarks should only be compared after aligning failure type, evidence assumption, intervention point, metric target, and action policy. We anchor the main synthesis in peer-reviewed or venue-linked work and use a 268-paper final adjudicated evidence map as trend and audit evidence. The result is a decision procedure for classifying robust-RAG papers, a small reliability pilot for pairwise comparability judgments, and an action-trace evaluation agenda for future benchmarks.

1 Introduction

Retrieval-augmented generation (RAG) combines a parametric language model with external evidence retrieved at inference time (Lewis et al., 2020). It is widely used for open-domain and document-grounded question answering because it can update knowledge without retraining and provide source-grounded answers. Yet retrieval also creates a new failure surface: evidence can be irrelevant, insufficient, conflicting, stale, unsupported by its citation, or maliciously injected.

Existing work studies these failures under many names: retrieval noise, distractor robustness, context faithfulness, knowledge conflict, corpus poisoning, indirect prompt injection, uncertainty estimation, selective generation, and abstention. We argue that these are instances of one broader problem: **robust RAG under unreliable evidence**. A robust RAG system should not only answer correctly when retrieval is helpful; it should estimate evidence reliability and choose an action when the evidence is unreliable: answer, abstain, re-retrieve, filter, verify, expose conflict, ask clarification, or escalate.

This paper is a *structured scoping map and agenda*, not an authoritative field census. We use an auditable corpus snapshot to organize the literature, but we do not claim PRISMA-level completeness or venue-complete coverage. Our main synthesis is anchored in peer-reviewed or venue-linked evidence; broader arXiv-heavy evidence is treated as trend and audit evidence. This positioning is deliberate: robust RAG is moving quickly, and strong claims should distinguish mature mechanisms from emerging preprint trends.

Why another RAG survey? General RAG surveys organize architectures, retrievers, generators, and applications (Gao et al., 2024; Sharma, 2025). Evaluation surveys organize datasets and metrics (Yu et al., 2024). Trustworthy and secure RAG surveys cover broad axes such as factuality, privacy, accountability, and attack surfaces (Zhou et al., 2024; Ni et al., 2025; Xu et al., 2026). These surveys provide useful coverage, but they do not give a concrete decision procedure for deciding when two robust-RAG papers are comparable. Our focus is narrower: when evidence is unreliable, what failure is present, where does the method intervene, what metric is being optimized, and what action should the system take?

The difference is not simply topical coverage. Prior surveys can tell a reader that retrieval, hallucination, security, and evaluation are important; our goal is to show why a reported robustness result cannot be interpreted without the evidence state and action policy. For example, a security survey may group poisoning and prompt injection as attacks, while an evaluation survey may group them by attack success rate. Under an evidence-reliability view, these are different games unless

the attacker's corpus access, query control, and retrieved evidence state are aligned. Similarly, a faithfulness survey may treat citation support as the central issue, while a calibration survey may focus on confidence; robust RAG requires asking whether the system should answer at all.

Contributions. We make three bounded contributions.

1. **A comparability decision procedure.** We specify how to align failure type, evidence assumption, intervention point, metric target, and action policy before comparing robust-RAG work.

2. **A tiered scoping evidence base.** We build a 1,399-candidate corpus snapshot, a 100-paper core reading set, a 326-paper venue-augmented scoping set, and a 268-paper final adjudicated evidence map; we separate peer-reviewed backbone claims from preprint-heavy trends.

3. **Paper-level re-analysis and agenda.** We apply the procedure to representative comparisons, identify invalid or partial comparisons, and state action-trace evaluation as a future benchmark agenda rather than a validated leaderboard.

2 Decision Procedure

The framework is a comparison procedure rather than only a taxonomy. Given a paper, the first question is what evidence property is unreliable: *relevance*, *sufficiency*, *consistency*, *provenance/freshness*, or *integrity*. The second question is where that failure comes from. A passage may be irrelevant because retrieval is noisy, contradictory because sources disagree, insufficient because the corpus lacks the answer, or corrupted because an attacker inserted it. These cases can all reduce answer accuracy, but they imply different repairs. A noisy passage can be filtered, an insufficient context may require re-retrieval or abstention, a conflict may require source comparison, and a poisoned document may require quarantine or traceback.

The third question is where the method intervenes. Some methods change the corpus or index, some change retrieval or reranking, some alter context assembly, some prompt or fine-tune the generator, and some act at the system-control layer. This matters because methods with the same goal may compose rather than compete. A retriever-side filter and a generator-side note-taking method can both improve robustness to irrelevant context, but one changes the evidence state and the other changes evidence use. Treating them as entries on the same leaderboard hides this difference.

The fourth question is what the metric actually measures. Exact match, retrieval recall, citation support, attack success, calibration error, selective risk, and latency are not interchangeable. A system can improve answer accuracy while worsening citation faithfulness; reduce attack success while losing utility; or improve calibration while still answering from insufficient evidence. The final question is action policy: what should the system do when evidence is unreliable? Answering, abstaining, re-retrieving, verifying, exposing conflict, and asking clarification are distinct outcomes.

This procedure prevents several common but misleading comparisons. A method that filters irrelevant passages, a method that trains a generator against noisy contexts, and a benchmark that measures negative rejection may all be called "noise robust," but they are not measuring the same object. Similarly, a poisoning defense with a bounded corruption guarantee and a prompt-injection defense may both lower attack success, yet one controls factual corruption and the other controls instruction hijacking. The unit of comparison must be a tuple:

$$(\text{failure}, \text{assumption}, \text{intervention}, \text{metric}, \text{action}).$$

If the tuple differs, the papers may be complementary but should not be ranked by one number.

The re-analysis in Appendix C applies this tuple to representative comparisons. IrrelCtx, RAAT, and RGB are all relevant to noise robustness, but one studies generator selectivity, one trains against noise distributions, and one decomposes benchmark abilities. Sufficient Context and Self-RAG both influence whether a system answers, but one estimates evidence sufficiency and the other learns a retrieval-and-critique policy. PoisonedRAG and certified defenses both concern corrupted evidence, but the former tests answer steering under poisoning while the latter assumes a bounded corruption model. These examples illustrate the central claim: many robust-RAG papers should be organized as complementary interventions rather than as competitors on a single leaderboard.

We also performed a limited operationalization check. From two independent human coding sources over the evidence map, we derived pairwise comparability labels for 75 paper pairs: 25 likely comparable pairs, 25 near-miss pairs sharing

a failure type but differing on another coordinate, and 25 cross-subfield negatives. The derived labels agree strongly (97.3%, $\kappa = 0.94$). This is not an independent expert user study, so we treat it only as feasibility evidence; a stronger validation package is provided for future expert annotation.

3 Evidence Base

The corpus begins with 1,399 deduplicated candidates and ends with a 268-paper final adjudicated evidence map. We also maintain a 100-paper core reading set and a 326-paper venue-augmented scoping set. The evidence map is not a field census: it is a curated and adjudicated map used to support the framework, identify trends, and make the corpus auditable. Details of screening, venue search, annotation, and adjudication are in Appendix E.

Two anonymized human sources independently coded 274 candidate evidence records before adjudication. Final adjudication removed six records from the evidence map. Agreement is strong for the fields that matter most to the framework: primary failure type ($\kappa = 0.96$), method family ($\kappa = 0.86$), intervention point ($\kappa = 0.92$), evaluation type ($\kappa = 0.93$), evidence strength ($\kappa = 0.95$), relevance level (weighted $\kappa = 0.80$), core-candidate status ($\kappa = 0.80$), and include-in-map decision ($\kappa = 0.85$). These results suggest that the taxonomy is usable by trained annotators, although they do not make the corpus venue-complete.

The map remains preprint-heavy: 42 final records are peer-reviewed/proceedings-linked and 222 are arXiv-only or emerging preprints, with a few other source strata. We therefore treat peer-reviewed or venue-linked work as the stable backbone and use the broader map for trend discovery. Under this stricter evidence stratum, five conclusions remain stable (Table 17): retrieved evidence can harm generation; natural conflict and adversarial poisoning require different assumptions; RAG security results require attacker-capability normalization; calibration and abstention are different action-control decisions; and answer accuracy alone is insufficient for robust-RAG evaluation.

How to read the map. The evidence map is not meant to estimate the true prevalence of every robust-RAG subproblem. Instead, it records how a curated, adjudicated corpus instantiates the comparison fields used in the framework. The key fields are not only topical labels, but the combination of evidence state, intervention point, metric target, evidence strength, and inclusion rationale. A preprint-heavy cluster can reveal emerging pressure points, while a smaller venue-linked cluster supports more stable mechanism-level claims. Readers should therefore use the map as a navigational and audit artifact: it helps locate comparable papers, identify under-specified comparisons, and decide which claims require stronger peer-reviewed support.

This distinction also shapes the role of the 100-paper core set. The core set anchors the main narrative and case studies, while the 268-paper adjudicated map supports broader trend inspection. We avoid treating map counts as community consensus. For example, if adversarial corruption appears dense in the map, that may reflect genuine research momentum, query design, and arXiv publication dynamics at the same time. The framework's stronger claim is not that one failure type is objectively most common; it is that any robust-RAG result should state which evidence failure it tests, what assumptions it makes, where it intervenes, and which system action it rewards.

4 Taxonomy and Reading Guide

The survey is organized around four layers. **Evidence-centric robustness** asks what enters the context: missing positives, irrelevant passages, hard negatives, stale evidence, conflicting sources, or poisoned documents. **Model-centric robustness** asks how the generator uses that evidence: whether it ignores support, over-trusts distractors, follows malicious instructions, or reconciles conflicting sources. **System-centric control** asks which action should be taken when evidence is unreliable: answer, verify, filter, retrieve again, abstain, clarify, or escalate. **Evaluation** asks whether benchmarks measure the right failure and reward the right action.

This four-layer taxonomy is useful only when paired with the comparison tuple above. A paper about retrieval noise and a paper about abstention may both improve final accuracy, but they operate on different states and should not be compared as if they optimize the same robustness property. Conversely, two papers in different sections may be comparable if their evidence state, intervention point, and metric target align. The taxonomy is therefore a reading guide rather than a rigid hierarchy: it helps locate a paper, while the decision procedure determines whether comparisons

are meaningful.

5 Condensed Literature Synthesis

Evidence construction: retrieval is not a neutral front end. Early robust-RAG work already shows that the retriever is not merely a recall module. Irrelevant-context studies demonstrate that a generator can be brittle even when the retrieved passage is obviously unrelated (Yoran et al., 2023), while noise-aware training studies show that different negative types produce different failure modes (Fang et al., 2024). Benchmark work such as RGB makes this explicit by separating noise robustness, negative rejection, information integration, and counterfactual robustness (Chen et al., 2024a). The stable lesson is that *relevant vs. irrelevant* is too coarse: the downstream effect of a passage depends on rank position, semantic hardness, redundancy, source quality, and whether gold evidence is also present. A retriever that maximizes recall can still degrade answer quality if it increases the density of plausible distractors.

This changes how retrieval-side methods should be compared. A reranker that demotes hard negatives, a sentence-level filter that removes local distractors, and a context compressor that preserves salient evidence all act before generation, but they solve different subproblems. Their failure cases also differ: aggressive filtering can remove weak but necessary multi-hop evidence; reranking can overfit to a particular generator or benchmark distribution; compression can delete qualifiers, timestamps, or uncertainty markers. These tradeoffs explain why downstream-utility metrics and evidence-sufficiency labels are needed in addition to retrieval recall.

Model-side evidence use: context availability is not context use. Once evidence enters the prompt, the generator must decide which passages to trust and how to reconcile them with parametric memory. Chain-of-Note style methods externalize intermediate evidence judgments before answering (Yu et al., 2023); conflict-aware methods explicitly model discrepancies between internal and retrieved facts (Zhang et al., 2025f; Wang et al., 2025b); and context-utilization benchmarks test whether models rely on gold evidence when distractors or conflicts are present. These methods reveal a second distinction: a passage can be available in context but not used, and a misleading passage can dominate even when support is available elsewhere.

Knowledge conflict is especially important because it sits between faithfulness, calibration, and retrieval quality. If retrieved evidence contradicts a model's parametric memory, blindly trusting retrieval can be wrong when the retrieved source is stale or poisoned; blindly trusting parametric memory can be wrong when the model is outdated. Robust RAG therefore needs source reliability, provenance, timestamp, and claim-level support, not only a preference for retrieved context. This is also why citation is not sufficient: a generated answer can cite a topically related passage that does not entail the claim, or cite the right document while using unsupported parametric knowledge. Claim-level verification remains expensive, but document-level citation is too weak for high-stakes evaluation.

Adversarial integrity: security results live inside threat models. RAG security work studies a different kind of unreliable evidence: evidence manipulated by an adaptive attacker. Corpus poisoning inserts malicious passages into the retrieval database; retriever poisoning manipulates dense similarity; indirect prompt injection hides instructions in retrievable documents; backdoor-style attacks rely on triggers; traceback methods identify malicious documents after a failure. These all corrupt the evidence channel, but their assumptions differ. A result that requires corpus write access and white-box retriever gradients should not be compared to a black-box attack that only controls a public document; a defense certified under a bounded corruption budget should not be read as protection against prompt-bearing retrieved text.

The main synthesis from the security backbone is therefore not a ranking of attacks or defenses, but a normalization rule. Every attack or defense result should report at least corpus write access, query control, retriever visibility, generator access, insertion budget, whether malicious evidence is retrieved, and whether the objective is answer steering, refusal induction, privacy extraction, traceback, or recovery. Without these fields, attack success rate is not portable. This is the clearest place where the comparability framework changes interpretation: many reported "RAG security" results are about different threat games.

System control: uncertainty must become action. Answerability and sufficient-context work asks whether the evidence is enough to answer (Joren et al., 2024). Adaptive retrieval asks whether external evidence is needed at all (Asai et al., 2024;

Yan et al., 2024). Calibration work asks whether confidence matches correctness or evidence sufficiency (Jang et al., 2025; Li et al., 2024b). These questions are related but not identical. A system can be confident but unsupported, uncertain but answerable after decomposition, or accurate but over-costly because it retrieves and verifies unnecessarily.

The action-control perspective therefore treats retrieval, abstention, verification, clarification, and escalation as policy decisions under cost and risk. The correct action depends on the evidence state: irrelevant evidence may require filtering, insufficient evidence may require re-retrieval or abstention, conflicting evidence may require source comparison or conflict disclosure, and poisoned evidence may require quarantine or traceback. This also explains why a single metric cannot summarize robust RAG. Accuracy, coverage, calibration, citation support, attack success, and latency define a tradeoff surface.

6 Critical Synthesis

Noise robustness is not one axis. Irrelevant-context filtering (Yoran et al., 2023), noise-aware training (Fang et al., 2024), Chain-of-Note style generator reasoning (Yu et al., 2023), and sufficient-context prediction (Joren et al., 2024) all address unreliable evidence, but they operate at different intervention points. Once distractor hardness, rank position, gold-evidence availability, and required action are aligned, many apparent competitions become complementary operating regions. The conclusion is not "method A is more robust to noise," but "method A repairs retrieval/context construction while method B changes generator reading or answer policy."

Conflict and poisoning should share annotations but not metrics. Knowledge-conflict work studies contradictions among parametric knowledge and retrieved sources (Jin et al., 2024b; Zhang et al., 2025f; Wang et al., 2025b). RAG poisoning work studies attacker-created evidence integrity failures (Zhong et al., 2023; Zou et al., 2025; Xue et al., 2024). Both need claim/source annotations, but their metrics differ: conflict work asks whether a system reconciles or exposes disagreement; poisoning work asks whether an attacker can steer answers under specified capabilities. Treating both as "counterfactual evidence" hides the threat model.

Security defenses transfer only inside a capability envelope. Certified defenses, reranking defenses, poisoned-document detectors, traceback methods, and prompt-injection defenses answer different questions. A certificate under a bounded corruption budget does not imply robustness to prompt-bearing documents. A traceback method may identify the poisoned document after failure without preventing user-facing harm. A prompt-injection filter may not address embedding-space poisoning. We therefore report prevention, demotion, detection, traceback, and recovery as different outcomes rather than one "defense success" score.

Calibration is not an action policy. Selective prediction and calibration ask whether confidence matches correctness (Geifman and El-Yaniv, 2017; Guo et al., 2017). In RAG, the target is richer: confidence may refer to answer correctness, evidence sufficiency, retrieval usefulness, citation support, or action utility. A model can be calibrated about answer correctness but still answer from insufficient evidence; an abstaining system can reduce risk while lowering coverage. Adaptive retrieval and abstention should therefore be evaluated as policies under cost and evidence state, not as confidence scores alone.

Benchmark rankings can be misleading. Benchmarks that report only final accuracy can reward systems that answer more often, retrieve more aggressively, or hide unresolved conflict. Benchmarks that report only attack success can hide utility loss or unrealistic attacker access. Benchmarks that report only faithfulness can ignore whether retrieval was necessary or sufficient. Robust-RAG evaluation should expose diagnostic sub-scores rather than one leaderboard scalar.

What changes after normalization? The re-analysis in Table 22 produces three kinds of outcomes. Some comparisons are *invalid*: for example, comparing a corpus-poisoning attack that assumes document insertion with a prompt-injection attack that assumes instruction-bearing retrieved text conflates factual corruption with instruction hijacking. Some comparisons are *partially valid*: two methods may share a failure type but differ in intervention point, such as a retriever-side filter and a generator-side note-taking method under irrelevant context. These should be compared as pipeline components, not substitutes. Some comparisons are *valid but under-specified*: two poison-

ing defenses may both evaluate attack success, but the comparison remains incomplete unless attacker budget, retriever visibility, and utility loss are reported.

This normalization changes research recommendations. For retrieval-noise work, the next useful benchmark is not simply harder negatives; it is a grid over negative type, rank position, support availability, and required action. For conflict work, the next useful evaluation is not only whether the model chooses the gold answer; it is whether the model detects source disagreement and explains why one source is preferred. For security work, the next useful leaderboard is not a single attack success rate; it is a threat-model-indexed table of prevention, detection, demotion, traceback, and recovery. For calibration work, the next useful metric is not only expected calibration error; it is selective risk conditioned on evidence sufficiency and retrieval cost.

The framework also identifies over-claimed results. A method can report strong robustness to irrelevant context while never testing semantically close hard negatives. A defense can report low attack success while assuming a corruption budget unavailable in open corpora. A benchmark can report high faithfulness while omitting cases where the correct action is to abstain or ask a clarification question. A long-context RAG system can include all evidence but still fail to select or cite the correct supporting span. These are not merely caveats; they determine whether a reported improvement transfers to another RAG pipeline.

Finally, the framework explains why combining modules is not automatically robust. A retriever filter, conflict resolver, verifier, and abstention policy can each improve its local metric while degrading another module's input. Filtering may remove the only weak multi-hop clue; verification may add latency that changes the acceptable action; abstention may reduce risk but also hide retriever failures that should trigger re-query. Robust RAG should therefore be evaluated as a controlled decision process over evidence states, not as a stack of independently optimized components.

Takeaways. The survey's main takeaway is that robust RAG is a coordination problem. Evidence-side methods decide what the model sees; model-side methods decide how evidence is used; security methods decide whether evidence can be trusted; and control methods decide whether the system should answer at all. Progress in one layer can fail to transfer when another layer changes. This is why the paper emphasizes alignment of failure, assumption, intervention, metric, and action rather than a new universal robustness score.

Open challenges. Three challenges stand out. First, benchmarks need compositional failures: noisy, conflicting, insufficient, and adversarial evidence should be combined rather than tested in isolation. Second, evaluation needs action labels: a correct answer, a justified abstention, and a useful clarification are different successes. Third, deployment needs cost-aware control: robust systems must decide when extra retrieval, verification, or human review is worth the latency and risk. These challenges motivate the action-trace agenda below.

7 Action-Trace Evaluation Agenda

The action-control view suggests that a robust-RAG benchmark should annotate not only answers but also evidence state and system action. Each item should specify the query, evidence state, required action, final answer target, cost budget, and–for adversarial cases–attacker capability. Actions include answer, abstain, re-retrieve, filter, verify, expose conflict, ask clarification, and quarantine evidence.

Appendix C gives the full blueprint, a 12-item synthetic illustration, and toy-policy scores. That pilot is not a benchmark release or main contribution; it only illustrates why answer-only scoring can hide action failures.

8 Challenges and Future Directions

Compositional evidence failures. Most current benchmarks isolate one stressor: irrelevant passages, conflicting passages, poisoned documents, or unanswerable questions. Real RAG systems encounter combinations. A query can be underspecified, retrieve a stale but plausible document, include a hard negative, and surface a malicious passage in the same context. The next generation of benchmarks should therefore compose failures rather than evaluate them one at a time. This is difficult because composition changes the correct action: a single hard negative may call for filtering, while hard negatives plus insufficient support may call for re-retrieval or abstention.

Evidence reliability estimation. Robust RAG needs better estimators of evidence state. Current

systems often estimate retrieval relevance, answer confidence, citation faithfulness, and safety with separate modules. These signals can disagree: a passage can be relevant but insufficient, sufficient but stale, or well cited but malicious. A useful reliability estimator should represent relevance, sufficiency, consistency, provenance, freshness, and integrity jointly. Such estimators should also expose uncertainty about their own judgments, because overconfident filtering can be as harmful as overconfident answering.

Transfer across retrievers and generators. Many robustness claims are evaluated with one retriever, one generator, one top-k, and one prompt format. This makes transfer hard to interpret. A defense that works for a dense retriever may fail for hybrid retrieval; a context-ordering method tuned for one generator may not transfer to another; a calibration policy can shift when model scale or instruction tuning changes. Future work should report robustness across retriever families, generator scales, context lengths, and prompt templates, or explicitly state that the claim is pipeline-specific.

From static evaluation to monitored systems. RAG corpora are mutable. Documents are updated, deleted, poisoned, de-duplicated, or re-indexed. A robust system should therefore log which evidence influenced each answer, monitor retriever drift, and support traceback when harmful outputs occur. This connects benchmark design to operational governance: a benchmark item is not only a query and answer, but also a corpus snapshot, provenance record, and action trace. Without these artifacts, robustness improvements can be hard to reproduce after the index changes.

Boundary areas. Multimodal, code, tool-using, and agentic RAG extend the same reliability problem but change the evidence object and action space. Images, tables, code files, tool outputs, and long-term memory can be evidence, instruction, or executable state. We treat these areas as boundaries rather than empirical backbone in this paper, but the same decision procedure should generalize if future work defines modality-specific evidence states and actions. This is a promising direction for follow-up surveys once the literature becomes less sparse.

9 Practical Guidance

For benchmark builders, the most important design choice is to annotate the evidence state before scoring the answer. A benchmark should specify which passages support the answer, which passages are distracting, which sources conflict, which documents are poisoned or stale, and whether the context is sufficient. Without these labels, an answer-only score cannot distinguish a model that reasoned from evidence from one that guessed correctly or relied on parametric memory. Benchmarks should also report action labels. If the correct behavior is abstention, clarification, or re-retrieval, then a fluent direct answer should be penalized even when it sounds plausible.

For method designers, the recommendation is to state the operating region explicitly. A filtering method should say which negatives it filters and whether it preserves weak multi-hop evidence. A calibration method should say whether it calibrates answer correctness, evidence sufficiency, retrieval usefulness, or abstention. A security defense should state whether it prevents attack, detects malicious evidence, demotes suspicious passages, traces a poisoned document, or recovers after failure. These distinctions make negative results more useful: a method that fails outside its operating region may still compose with another method that covers the missing layer.

For system builders, robust RAG should be treated as risk-adaptive control rather than a fixed retrieve-then-answer pipeline. Low-risk FAQ systems may only need lightweight filtering and citation checks. Open-web or user-editable corpora require provenance tracking, ingestion filters, and poisoned-document monitoring. High-stakes domains require answerability checks, conflict disclosure, selective abstention, and escalation paths. The framework does not prescribe one universal stack; it provides a vocabulary for matching controls to evidence risks.

10 Limitations and Ethics

This paper is a scoping map, not a field census. The corpus is open-PDF based, not venue-complete, and remains preprint-skewed. The venue-oriented pass and follow-up triage improve auditability but do not replace manual proceedings review over ACL, EMNLP, NAACL, COLING, SIGIR, WWW, WSDM, KDD, USENIX Security, CCS, NDSS, ICLR, and NeurIPS. The 268-paper map should

therefore be read as a final adjudicated map of our corpus, not as field-wide prevalence.

The scope is text/document-grounded RAG. Multimodal, agentic, code, tool-use, and persistent-memory RAG are boundary areas because their evidence objects and action spaces differ. They appear in the appendix and future directions but are not the empirical backbone.

The survey discusses poisoning, prompt injection, backdoors, opinion manipulation, and privacy extraction at the level of threat model, failure mechanism, evaluation signal, and defense implication. We do not provide operational payloads or deployable attack instructions. Evaluation artifacts should use synthetic or sandboxed poisoned documents and avoid private data unless consent and retention controls are explicit.

11 Conclusion

Robust RAG is not only a question of whether retrieval improves an answer. It is a question of whether the system can judge evidence reliability and choose the right action under uncertainty. The peer-reviewed and venue-linked backbone supports the mechanism-level conclusion that evidence failures, attacker assumptions, calibration targets, and action policies must be aligned before methods or benchmarks are compared. The broader adjudicated evidence map is useful as trend and audit evidence, not as a census. Future robust-RAG benchmarks should therefore evaluate answer correctness, evidence support, action quality, cost, and threat assumptions together.

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A Extended Survey by Failure Mode and Intervention

This appendix expands the compact synthesis in the main paper. Its purpose is not to add a citation dump, but to explain how the broader evidence map is organized by failure mode, intervention point, and evaluation target. We group papers by their primary role in the framework; many papers naturally span multiple categories.

A.1 Evidence-Centric Robustness: Noisy Retrieval and Distraction

The first branch of robust RAG studies what enters the context. Early work on irrelevant context established that retrieval errors do not merely lower recall; they can actively redirect generation (Yoran et al., 2023). Noise-aware training and robust RALM work then separated random negatives, hard negatives, counterfactual passages, and noisy retrieved evidence as different stressors rather than one generic perturbation (Fang et al., 2024; Park and Lee, 2024; Ding et al., 2025). Benchmark papers such as RGB and URAG made this decomposition operational by testing negative rejection, information integration, counterfactual evidence, and uncertainty under retrieval noise (Chen et al., 2024a; Nguyen et al., 2026). These works motivate

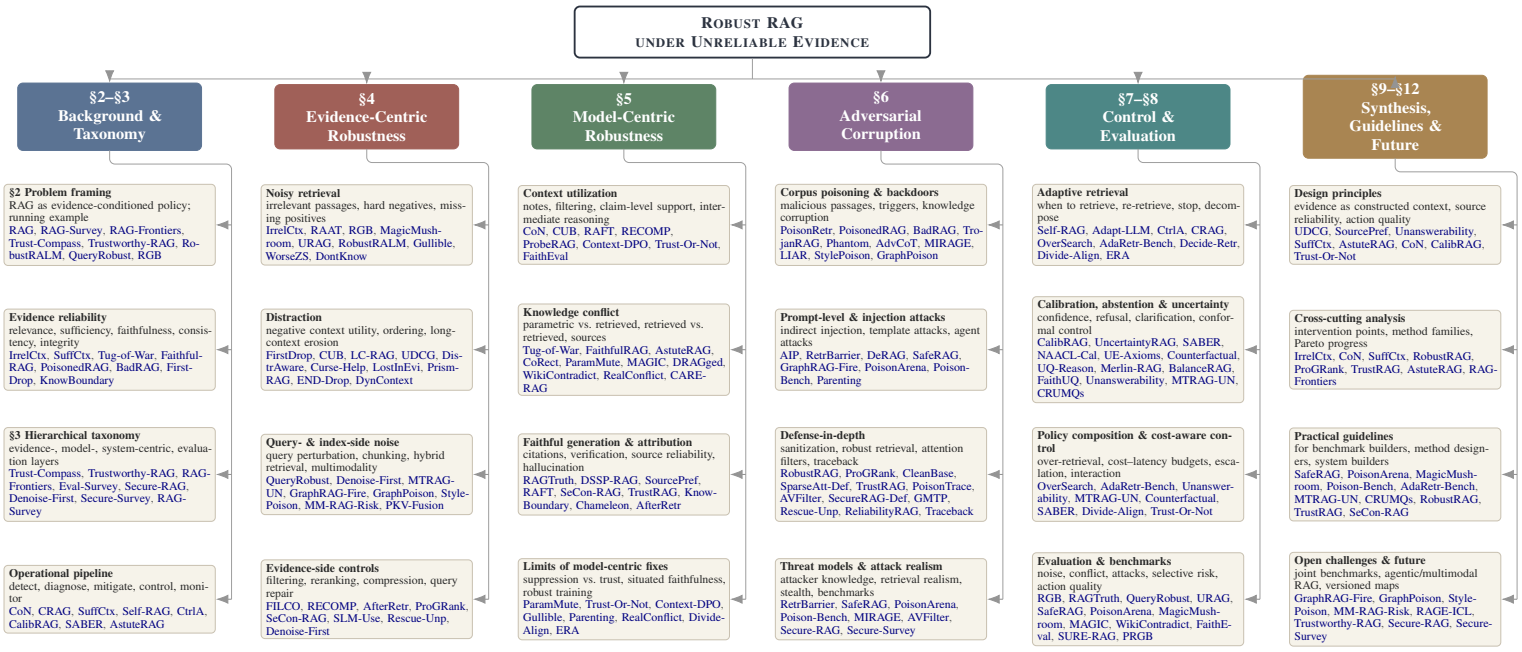


Figure 1: **Outline of this survey.** A top-down view of the paper structure. The root problem is robust RAG under unreliable evidence; the survey then moves from background and taxonomy to evidence-centric, model-centric, adversarial, system-control, evaluation, synthesis, and future-facing guidelines. Each leaf lists representative cited work using compact aliases.

the main-paper claim that "noise robustness" is not a single axis.

Distraction work focuses on negative marginal utility: a passage is harmful not simply because it is irrelevant, but because it changes the answer, confidence, attribution, or reasoning path. Long-context and context-utility studies show that more context can increase the probability of including plausible but misleading passages (Gao et al., 2026b; Jin et al., 2024a; Hagström et al., 2025). Utility-oriented retrieval and downstream-aware metrics therefore ask whether a passage helps the generator rather than whether it only matches the query (Trappolini et al., 2025; Zhu et al., 2024; Dai et al., 2026; Niu et al., 2026; Gao et al., 2025). Context compression and filtering methods try to remove irrelevant spans or documents while preserving evidence needed for multi-hop reasoning (Wang et al., 2023b; Yu et al., 2023; Joren et al., 2024; Hong et al., 2023; Cho et al., 2024; Zeng et al., 2025).

The later evidence-centric literature broadens the source of noise beyond the top-k ranking. Query-level robustness work studies sensitivity to paraphrase and wording changes (Percin et al., 2025); long-context RAG and context-position studies examine how ordering and context length alter evidence use (Kim et al., 2026; Gabín et al., 2026; Pandey, 2026; Wang et al., 2025e; Chen et al., 2025a; Liu et al., 2025e). Other papers study context construction, compression, and retrieval defects as pipeline-level problems rather than isolated retriever errors (Chen et al., 2025b; Yang et al., 2025; Kachuee et al., 2025; Chakraborty et al., 2025; Oh et al., 2025; Tan et al., 2025; Kim, 2025). More recent systems and benchmarks extend this line with stress tests for retrieval defects, context utility, and robust fine-tuning (Jin et al., 2025; Iratni et al., 2025; Sui et al., 2025; Qiao et al., 2024; Wang et al., 2024a; Zhang et al., 2024a; Qiu et al., 2024; Park et al., 2024; Abdumalikov et al., 2024; Kim et al., 2024; Xu et al., 2023). The common lesson is that retrieval should be evaluated as context construction.

A.2 Model-Centric Evidence Use: Conflict, Faithfulness, and Attribution

Once evidence enters the prompt, robustness depends on how the generator uses it. Knowledge-conflict papers study contradictions between parametric memory and retrieved evidence, or among retrieved sources (Jin et al., 2024b; Zhang et al., 2025f; Wang et al., 2025b; Xie et al., 2023). These papers show that a model can over-trust retrieved evidence, resist correct retrieval because of parametric priors, or silently choose one source without exposing disagreement. Conflict-aware methods therefore localize discrepancies, compare internal and external knowledge, or introduce source-aware

reasoning before generation (Ma et al., 2026; Zhou et al., 2025a; Si et al., 2025; Huang et al., 2025a; Lee et al., 2025; Cattan et al., 2025; Hou et al., 2024; Huang et al., 2024).

Faithfulness and attribution papers study whether generated claims are supported by evidence. RAGTruth and related grounding work provide fine-grained hallucination or support labels, while source-preference and entity-conflict studies examine whether models cite and trust the right evidence (Niu et al., 2024; Pham et al., 2024; Kortukov et al., 2024; Zhang et al., 2024b; Schuster et al., 2026). These papers connect robust RAG to citation verification: document-level citation is often too coarse, because a document can contain multiple claims, qualifiers, and contradictions. Claim-level or sentence-level attribution is more informative but increases annotation and verification cost (Zhao et al., 2026a; Shin et al., 2026; Lin et al., 2025b; Choi et al., 2025b; Wu et al., 2024; Dai et al., 2025a; Ge et al., 2025).

The broader conflict and faithfulness map includes work on source reliability, contextual faithfulness, real-document conflicts, and hidden-state or logit interventions (Jin et al., 2024c; Wang et al., 2023a; Li et al., 2026b; Shihab et al., 2026; Ju et al., 2026; Liu et al., 2025d; Yan et al., 2025; Huo et al., 2025). Newer papers extend conflict analysis to richer benchmark settings, source-aware generation, and practical RAG pipelines (Wu et al., 2025b; Long et al., 2025; Jia et al., 2025b; Wang et al., 2025d; Xu et al., 2024; Li et al., 2024a; Yang et al., 2024; Marjanović et al., 2024; Tan et al., 2024a; Zhang et al., 2023). Across these papers, the main-paper distinction between context availability and context use recurs: evidence can be present but ignored, cited but unsupported, or correct but less trusted than a stale parametric belief.

A.3 Adversarial Evidence Corruption and Security

Adversarial RAG work asks what happens when evidence unreliability is created by an adaptive attacker. Corpus and retrieval poisoning papers show that small numbers of malicious passages can manipulate dense retrieval or steer final answers (Zhong et al., 2023; Zou et al., 2025; Xue et al., 2024; Geng et al., 2025; Zhang et al., 2025a). Certified and aggregation-based defenses study robustness under bounded corruption assumptions (Xiang et al., 2024; Cheng et al., 2025; Kim et al., 2025a), while reranking and detection defenses use gradient, attention, clustering, or anomaly signals to demote suspicious evidence (Yin et al., 2026; Jin et al., 2026; Dekel et al., 2026; Choudhary et al., 2025; Si et al., 2025).

Prompt-injection and backdoor work targets a different interface. Instead of inserting false facts, the attacker may insert instructions that the model treats as commands, or triggers that activate malicious behavior through retrieved contexts (Chaturvedi et al., 2025; Chang et al., 2026; Cheng et al., 2024; Chaudhari et al., 2024; Song et al., 2026; Kim et al., 2025b). These attacks require retrieval realism: a malicious document must be retrieved under plausible queries, not only under attacker-crafted prompts. This is why the main paper separates factual poisoning, indirect prompt injection, backdoor triggers, and prompt-template attacks as different threat games even when they all appear as retrieved text.

The security evidence map is broad and fast-moving. It includes poisoning benchmarks and arenas (Liang et al., 2025b; Chen et al., 2025d; Zhang et al., 2025e; Tan et al., 2024c), trace-back and forensics (Zhang et al., 2025d,c), privacy and membership inference (Li et al., 2026a; Gao et al., 2026a; Song et al., 2025b; Shereen et al., 2025), opinion or fairness manipulation (Chen et al., 2025h; Bagwe et al., 2025), and graph, multimodal, or reasoning-style poisoning surfaces (Liang et al., 2025a; Wen et al., 2025a; Zhou and Wang, 2025). Many defenses and attacks are preprint-heavy (Pathmanathan et al., 2025; Jiao et al., 2025; Shao et al., 2025; Li et al., 1986; Wang et al., 2025c; Chen et al., 2025f, 2024e; Wu et al., 2025c; Zhou et al., 2025b; Wu and Saxena, 2025; Gong et al., 2026; Hu et al., 2026; Xian et al., 2025; Shah and Ozgur, 2026; Gowda, 2026; Yu et al., 2026; Zhang et al., 2026a; Korn, 2026; Yao et al., 2025; Dai et al., 2025b; Sui, 2025; Xian et al., 2024; Yang et al., 2026; Choi et al., 2025a; Zhang et al., 2025b; Gong et al., 2025; Dhar et al., 2025; Liu et al., 2025f; Wen et al., 2025b; Wang and Yu, 2025; Song et al., 2025a; Stefano et al., 2024; Wang et al., 2024b; Pasquini et al., 2024; Clop and Teglia, 2024; Wang et al., 2025a; Li et al., 2025c; Xi et al., 2025; Zhao et al., 2025; Chang et al., 2025; Ha et al., 2025; Suo et al., 2025; Tan et al., 2024b; Levi, 2026; Mu et al., 2026; Wang et al., 2026b; Liu et al., 2026c; Wang et al., 2026a; Lee et al., 2026; Xiao et al., 2026; Chen et al., 2026a; Guo and Wei, 2026; Edemacu and Shokri, 2026; Zhang et al., 2026b; Rahman and Alouani, 2026; Qi et al.,

2026; Habler et al., 2026; Lyu et al., 2026; Datta et al., 2026; Castagnaro et al., 2025; Chen et al., 2025g; Liu et al., 2025c; Jia et al., 2025a; Zhu et al., 2025; Srivastava and He, 2025; Shang et al., 2025; Abdelnabi et al., 2025; Ye et al., 2025; Zuo et al., 2025; Li et al., 2025b; Lin et al., 2025a; Zhou et al., 2025c; Edemacu et al., 2025; Wang et al., 2025f; Chen et al., 2024c; Hu et al., 2024; Abdelnabi et al., 2024; Chen et al., 2024d; Yu et al., 2025). The main use of this large set is not to rank attacks, but to motivate capability-normalized reporting.

A.4 System Control: Uncertainty, Calibration, and Abstention

System-control work converts evidence judgments into actions. Answerability and sufficient-context papers ask whether the retrieved context contains enough evidence to answer (Joren et al., 2024; Peng et al., 2024; Rosenthal et al., 2026; Liu et al., 2025a). Uncertainty and calibration work asks whether confidence reflects answer correctness, evidence sufficiency, retrieval usefulness, or selective risk (Jang et al., 2025; Li et al., 2024b; Nguyen et al., 2026; Liu et al., 2026b; Soudani et al., 2025a; Qiu et al., 2026). Adaptive retrieval work asks when the system should retrieve, retrieve again, or answer from internal knowledge (Asai et al., 2024; Labruna et al., 2024; Yan et al., 2024; Liu et al., 2025b; Moskvoretskii et al., 2025).

The control literature highlights a recurring failure: confidence is not the same as evidence sufficiency. A model can be confident in an unsupported answer, or uncertain even when evidence is present but hard to integrate. Trust-or-abstain and self-aware RAG methods therefore model whether to answer, abstain, retrieve more, or ask for clarification (Zhu et al., 2026; Xie et al., 2026; Chen et al., 2024b; Soudani et al., 2025b; Sun et al., 2025; Chen et al., 2026c; Zhao et al., 2026b; Deiseroth et al., 2025; Jia et al., 2026; Fadeeva et al., 2025; Huang et al., 2025b). Multi-turn and underspecified-query benchmarks add another layer: the correct action may be clarification before retrieval, not direct answering (Rosenthal et al., 2026; Bazarova et al., 2026; Chen et al., 2026b).

Recent work extends control to cost-aware retrieval, conformal or set-valued prediction, risk-sensitive abstention, and adaptive policies over multiple retrieval steps (Zhou et al., 2026; Sinha, 2025; Wu et al., 2025a; Zubkova et al., 2025; Wang et al., 2025g; Chen et al., 2025c; Li et al., 2025a; Marina et al., 2025; Foo et al., 2025; Shankar et al., 2025; Huang et al., 2025c; Min et al., 1913). These papers motivate the main paper's action-control view: robust RAG is not only about reducing errors, but about choosing the right response under uncertainty and cost.

A.5 Evaluation, Benchmarks, and Survey Context

Evaluation papers provide the diagnostic infrastructure for robust RAG. RGB decomposes RAG reliability into multiple abilities (Chen et al., 2024a); RAGTruth provides fine-grained hallucination annotations (Niu et al., 2024); query-robustness and context-utility work stress retrieval interfaces (Percin et al., 2025; Trappolini et al., 2025); security benchmarks and arenas evaluate poisoning and defense assumptions (Liang et al., 2025b; Chen et al., 2025d; Tan et al., 2024c); and uncertainty benchmarks evaluate whether systems know when their answers are reliable (Nguyen et al., 2026; Qiu et al., 2026). Survey papers on RAG, trustworthy RAG, secure RAG, and evaluation provide the broader landscape (Gao et al., 2024; Sharma, 2025; Yu et al., 2024; Zhou et al., 2024; Ni et al., 2025; Xu et al., 2026; Mu et al., 2026; Liu et al., 2026a).

What is still missing is joint evaluation. Existing benchmarks often isolate one failure mode: noise, conflict, poisoning, or unanswerability. Real systems face combinations. A benchmark that includes a stale source, a hard negative, insufficient evidence, and a malicious document will test a different capability from any single-axis benchmark. This motivates the action-trace blueprint: robust RAG evaluation should score evidence state, answer correctness, action choice, cost, and threat assumptions together.

B Detailed Background and Taxonomy

C Evaluation Details

D Exploratory Map Statistics

E Detailed Audit Artifacts

F Core and Evidence Map Samples

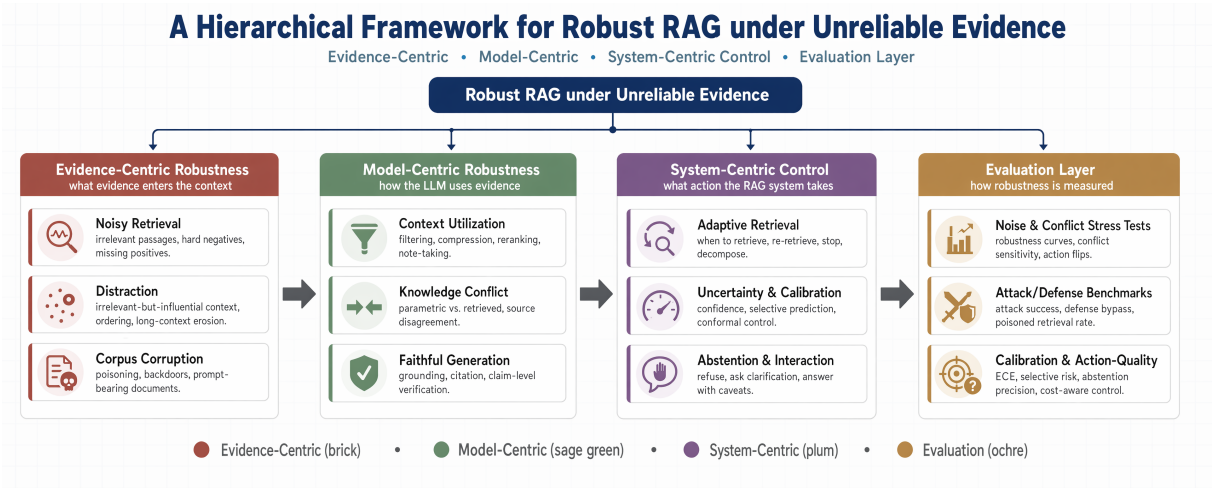


Figure 2: Research landscape map for robust RAG. The figure organizes the literature by where robustness problems and methods sit in the RAG stack: evidence-centric work controls the reliability of retrieved inputs, model-centric work controls how the generator uses evidence, system-centric work studies control decisions, and evaluation work measures robustness and action quality.

Step	Question to ask	Decision rule	Common boundary case
1. Evidence state	What is unreliable: relevance, sufficiency, consistency, provenance/freshness, or integrity?	Code the failure by the evidence property being tested, not by the paper's broad motivation.	A hard negative is noisy retrieval until the paper measures how it changes generation, then distraction becomes relevant.
2. Source of failure	Is the failure benign, naturally occurring, benchmark-constructed, or attacker-created?	Natural contradiction is knowledge conflict; attacker-created contradiction is adversarial corruption.	Poisoned facts and conflicting facts can look identical in context but require different evaluation assumptions.
3. Intervention point	Where does the method first change system behavior?	Corpus, retriever, context assembly, generator, system control, or evaluation.	Context filtering and generator prompting may both improve robustness but are not direct competitors.
4. Metric target	What is being optimized or measured?	Separate answer accuracy, evidence support, citation faithfulness, attack success, calibration, selective risk, and cost.	A method can improve EM while worsening citation support or increasing unsafe coverage.
5. Action policy	What should the system do when evidence is unreliable?	Answer, abstain, re-retrieve, filter, verify, expose conflict, ask clarification, or escalate.	Abstention after one weak retrieval may be premature if re-retrieval is cheap.
6. Comparability test	Are all five fields aligned across papers?	Compare only if failure state, assumptions, intervention point, metric, and action policy match.	Otherwise report complementary operating regions rather than a single ranking.

Table 1: Decision procedure for applying the evidence-reliability/action-control framework to a paper or benchmark.

Prior view	What changes under this framework
RAG robustness as a model or retrieval module property	Robustness is a tuple: failure, assumption, intervention, metric, and action. A method can improve one coordinate while leaving another failure untouched.
RAG security as attack/defense taxonomy	Attack/defense claims are only comparable after normalizing corpus write access, query control, retriever visibility, budget, and output target.
RAG evaluation as answer or faithfulness score	Answer accuracy, citation support, evidence sufficiency, calibration, and action quality can diverge and need separate reporting.

Table 2: Main-text delta relative to adjacent RAG surveys. The contribution is not broader coverage, but a different comparison rule.

Ambiguous boundary	Coding rule	Example implication	Why it matters
Noisy retrieval vs. distraction	Code by the measured effect: retrieval imperfection alone is noise; harmful influence on generation is distraction.	Random negatives and hard distractors should not be pooled into one score.	Explains why recall gains can reduce answer quality.
Conflict vs. adversarial corruption	Code by source of contradiction: naturally occurring or diagnostic contradiction is conflict; attacker-created contradiction is corruption.	PoisonedRAG and FaithfulRAG should not be compared by the same metric.	Separates reliability failures from adaptive threat models.
Uncertainty vs. calibration/control	Uncertainty estimates belief state; calibration/control decides action from that belief.	URAG and Self-RAG answer different questions.	Prevents treating abstention as just another confidence score.
Retriever vs. generator interventions	Code by first intervention point that changes system behavior.	FILCO filters context; Chain-of-Note changes generator reasoning.	Reveals why methods with similar goals may compose rather than compete.
Benchmark vs. method	Code benchmark if the reusable artifact is the main contribution; otherwise code the method.	RGB is benchmark-first; RAAT is training-first.	Clarifies whether a paper contributes evidence or an intervention.

Table 3: Taxonomy decision rules used to handle ambiguous cases. These rules turn the taxonomy from a pipeline decomposition into a comparison framework: methods are comparable only when their failure source, intervention point, and evaluation target align.

Stage	Question	Typical signals	Representative actions
Detect	Is the current evidence unreliable?	Low retrieval confidence, high contradiction, low sufficient-context score, suspicious document patterns	Flag query/evidence, route to stronger checks, log risk state
Diagnose	What kind of unreliability is present?	Irrelevance, distraction, conflict, poisoning, ambiguity, unanswerability, stale evidence	Assign failure type; separate benign noise from adversarial risk
Mitigate	Can the evidence be repaired before answering?	Alternative retrieval results, filter scores, reranker utility, provenance, source agreement	Re-rank, filter, compress, re-query, retrieve from trusted sources, remove malicious passages
Control	What should the system do now?	Calibrated confidence, selective risk, cost budget, user risk tolerance	Answer, abstain, ask clarification, expose conflict, escalate to human review
Monitor	Did the system fail or get attacked?	User feedback, traceback signals, poisoned-document clusters, drift over time	Update index, quarantine sources, revise prompts, retrain filters, update audit logs

Table 4: An operational pipeline for robust RAG. The pipeline turns the survey taxonomy into actionable system design steps.

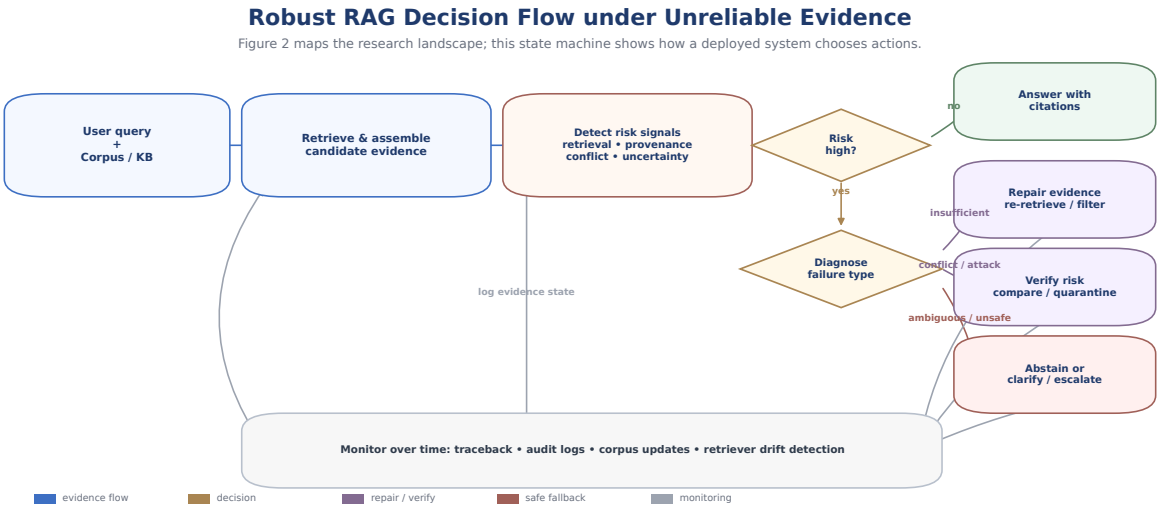


Figure 3: Operational decision flow for robust RAG. Unlike the taxonomy in Figure 2, this state-machine view describes the branching actions a deployed system can take when unreliable evidence is encountered: answer, repair evidence, verify risk, abstain, clarify, escalate, or monitor.

Theme	Representative work	Core idea	What it reveals for robust RAG
Irrelevant context	IrrelCtx	Analyze and mitigate irrelevant retrieved context with filtering and robust training	Even answerable questions degrade when irrelevant passages enter the context; robustness is not equivalent to retrieval recall.
Retrieval-noise training	RAAT	Categorize retrieval noise and train with adaptive adversarial objectives	Noise types should be separated: irrelevant, relevant-but-insufficient, and counterfactual evidence have different effects.
Generator note-taking	CoN	Generate per-passage notes before answering	Robust generation benefits from intermediate relevance judgments rather than direct answer generation.
Fine-grained noise benchmark	MagicMushroom	Configurable benchmark for distracting, low-quality, inconsequential, and irrelevant noise	"Noise" is not monolithic; evaluation should vary type, amount, and placement of bad evidence.
Nonlinear distraction	FirstDrop	Study disproportionate effect of small amounts of misleading information	A few strong distractors can dominate long-context reasoning, so average noise rate is insufficient.
Long-context RAG	LC-RAG	Analyze long-input RAG and hard negatives	More retrieved context can hurt unless ordering, filtering, or training adapts to long-context failure modes.
Sentence-level filtering	FILCO	Filter retrieved context before generation	Filtering can be more fine-grained than document reranking and can remove local distractors inside otherwise useful passages.
Retrieval evaluation	UDCG	Utility-aware retrieval metric for LLMs	RAG retrieval should be evaluated by downstream utility, including negative utility from distracting passages.

Table 5: Representative evidence-centric work on noisy retrieval and distraction.

Problem	Representative work	Mechanism	Limitation or open issue
Parametric–retrieved conflict	Tug-of-War	Construct and analyze conflicts between internal and external knowledge	Conflict behavior depends on model, prompt, evidence coherence, and task framing.
Fact-level conflict modeling	FaithfulRAG	Extract parametric self-facts, align them with retrieved facts, and reason before answering	Requires reliable fact extraction and alignment; complex claims remain difficult.
Source-aware consolidation	AstuteRAG	Elicit internal knowledge and consolidate it with retrieved sources using reliability comparison	Training-free but still depends on LLM self-assessment and source-quality cues.
Hidden-state rectification	CoRect	Locate layers where parametric suppression occurs and contrast contextual/non-contextual logits	Promising for model-centric control, but model internals and transferability remain open.
Context utilization benchmarks	CUB	Test gold, conflicting, and irrelevant contexts across models	Benchmarks reveal whether a model uses context, but not always why a model chooses one source.
Real-document conflict	RealConflict	Study behavior under context-memory conflicts with real documents	Realistic evidence can be messier than synthetic conflicts and harder to label.
Source preference	SourcePref	Analyze which sources win under conflict	Source preference may encode bias, recency effects, or authority priors.

Table 6: Representative model-centric work on knowledge conflict and contextual faithfulness.

Attack/defense surface	Representative work	Threat or defense	Implication
Retriever poisoning	PoisonRetr	Inject adversarial passages to manipulate dense retrievers	Retrieval security can fail before generation begins.
RAG knowledge corruption	PoisonedRAG	Craft malicious passages that steer generated answers	End-to-end RAG needs corpus integrity and generation safeguards.
Backdoor attacks	TrojanRAG; Phantom	Triggered retrieval/generation behavior via poisoned contexts	RAG can act as a backdoor carrier without modifying model weights.
Prompt template attacks	AIP	Weaponize shared instructional prompts to steer retrieval and generation	Prompt templates are part of the attack surface, not just implementation details.
Indirect prompt injection	RetrBarrier	Optimize trigger fragments so malicious instructions are retrieved in the wild	Real attacks must overcome the retrieval barrier, not merely place malicious text in a corpus.
Certified defense	RobustRAG	Isolate-and-aggregate answers for robustness certificates	Certification gives stronger guarantees but typically assumes specific attack budgets.
Reranking defense	ProGRank	Use probe-gradient instability to demote poisoned passages	Defense can be inserted retriever-side without retraining the LLM.
Attention defense	AVFilter	Detect poisoning via normalized attention variance	Internal attention signals may expose poisoned evidence, but adaptivity remains a concern.

Table 7: Representative adversarial corruption attacks and defenses.

Threat class	Corpus	Query	Retr.	Gen.	Objective	Boundary
Dense-retriever poisoning (Zhong et al., 2023)	write	train set	white/gray	no	retrieve bad passages	Not always end-to-end.
RAG knowledge corruption (Zou et al., 2025; Xue et al., 2024)	write	target topic	black/gray	black	answer steering	Stealth/dynamics vary.
Backdoor-style RAG (Cheng et al., 2024; Chaudhari et al., 2024)	write	trigger	gray	black	latent behavior	Normal queries may pass.
Indirect prompt injection (Chaturvedi et al., 2025; Chang et al., 2026)	write/share	natural	black	black	instruction hijack	Must overcome retrieval.
Opinion/fairness manipulation (Chen et al., 2025h; Bagwe et al., 2025)	write	group/topic	black/gray	black	bias steering	QA metrics miss social harm.
Defense setting (Xiang et al., 2024; Yin et al., 2026; Choudhary et al., 2025; Zhang et al., 2025c)	bounded	varies	assumed	black/gray	detect/demote/trace	Depends on budget/visibility.

Table 8: Attacker-capability matrix for adversarial RAG. The same "attack success rate" can mean different things depending on corpus write access, query control, retriever visibility, generator access, and objective.

Comparison	Why it looks comparable	Hidden mismatch	Metric pitfall	Normalized reading
PoisonedRAG vs. dense-retriever poisoning	Both insert malicious evidence into a retrieval corpus.	One evaluates end-to-end answer steering; the other may evaluate retriever ranking before generation.	Retrieval success and answer attack success are not the same target.	Compare only after recording whether poisoned evidence is retrieved and whether the generator changes the final answer.
Certified RAG defense vs. prompt-injection defense	Both are RAG defenses against corrupted retrieved content.	Certification assumes a bounded number of corrupted passages; prompt injection exploits instruction following rather than only false facts.	A certificate under bounded corruption may not imply instruction-hijack resistance.	Treat certification, instruction isolation, and document sanitization as complementary defenses.
Traceback vs. prevention	Both reduce harm after poisoning.	Traceback identifies influential poisoned documents after failure; prevention attempts to block harmful output before response.	Lower post-hoc traceback error does not equal lower user-facing attack success.	Report prevention, detection, demotion, traceback, and recovery as separate outcome variables.
Backdoor trigger vs. natural query attack	Both can produce targeted malicious behavior.	Trigger attacks may rely on rare phrases; natural-query attacks must survive ordinary retrieval and ranking.	Attack success under attacker-crafted triggers overstates risk under natural user queries.	Require query-control and trigger-frequency fields before comparing attack strength.

Table 9: Concrete examples of threat-model normalization. These cases show how reported attack or defense success can reverse when attacker access, retrieval realism, and evaluation target differ.

Control/evaluation question	Representative work	Signal or decision	Why it matters
When to retrieve	Self-RAG; Adapt-LLM; CtrlA	Reflection tokens, special retrieval tokens, or representation-based confidence/honesty directions	Retrieval is an action with costs and risks, not a fixed preprocessing step.
When retrieval is weak	CRAG; SuffCtx	Retrieval evaluator or sufficient-context signal	A system should repair or abstain rather than answer from insufficient evidence.
Calibration	CalibRAG; UncertaintyRAG; NAACL-Cal	Decision calibration, span uncertainty, verbal confidence calibration	Robustness requires confidence to reflect both model and evidence uncertainty.
Unanswerability	Unanswerability; MTRAG-UN	Unanswerable and underspecified RAG benchmarks	Real users often ask questions the corpus cannot answer; refusal and clarification need direct evaluation.
RAG capabilities	RGB	Noise robustness, negative rejection, integration, counterfactual robustness	Multi-axis evaluation exposes failures hidden by answer accuracy.
Hallucination detection	RAGTruth	Word-level hallucination annotations for RAG outputs	Fine-grained labels support detection and attribution beyond answer-level scoring.
Query robustness	QueryRobust	Perturb query wording and measure pipeline sensitivity	Robustness must include the query-to-retrieval interface.

Table 10: Representative work on system-centric control and evaluation.

Lineage	Classical question	RAG-specific version	Why it matters for robust RAG
Selective prediction	When should a model abstain to control risk?	Abstain when retrieved evidence is insufficient, conflicting, stale, or corrupted.	Selective risk must condition on evidence state, not only answer confidence.
Calibration	Does confidence match correctness?	Does confidence match answer correctness, evidence sufficiency, citation support, and retrieval usefulness?	A calibrated answer probability can still hide unreliable evidence.
Active clarification	When should the system ask a user for more information?	Ask clarification before retrieval when the query is ambiguous or underspecified.	Clarification can be the correct action even when retrieval could produce a plausible answer.
Cost-sensitive retrieval	When is additional computation worth it?	Decide whether to retrieve, re-retrieve, verify, rerank, or escalate under latency/cost budgets.	Robustness gains should be reported with action cost and coverage, not only accuracy.
Decision-theoretic control	Which action minimizes expected risk?	Choose among answer, abstain, expose conflict, cite uncertainty, retrieve again, or quarantine evidence.	Action quality is the bridge between evidence reliability estimates and user-facing behavior.

Table 11: Action-control lineage. The framework imports ideas from selective prediction, calibration, clarification, and cost-sensitive decision-making, but makes retrieved evidence state the object being controlled.

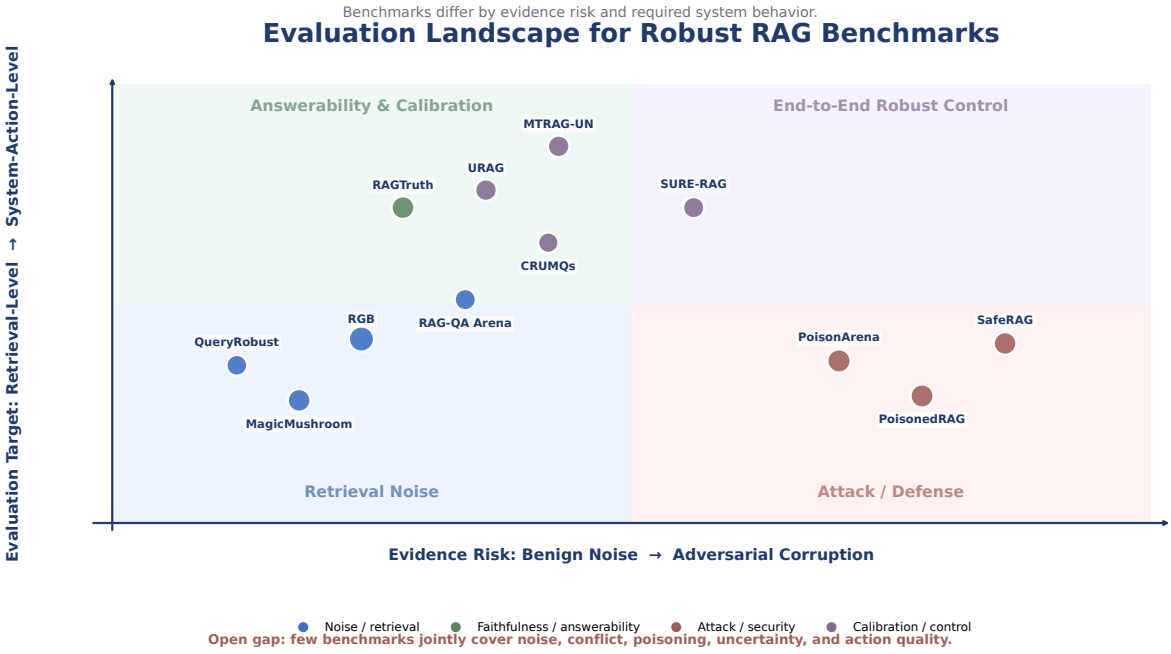


Figure 4: Evaluation landscape for robust RAG benchmarks. Existing benchmarks differ in the type of evidence risk they simulate and whether they evaluate retrieval-level behavior, generation-level behavior, or system-action-level robustness.

Benchmark	Risk focus	Task / data	Perturbation or threat	Metrics	Boundary
RGB (Chen et al., 2024a)	noise, rejection, integration, counterfactuality	English/Chinese QA	noisy passages, false evidence, negative evidence	accuracy, ability-specific scores	Broad diagnostic benchmark but not adaptive attack-oriented.
RAGTruth (Niu et al., 2024)	hallucination / faithfulness	QA, data-to-text, summarization	naturally generated unsupported spans	word-level hallucination labels	Strong annotation granularity; less focused on retrieval-side threat models.
Sufficient Context (Joren et al., 2024)	answerability	QA with context sufficiency labels	insufficient retrieved evidence	selective generation / abstention	Captures answerability but not adversarial corruption.
MagicMushroom (Zhang et al., 2025g)	retrieval noise	configurable RAG stress tests	distracting, low-quality, inconsequential, irrelevant noise	robustness under noise types	Fine-grained noise; limited conflict/security coverage.
URAG (Nguyen et al., 2026)	uncertainty	RAG as MCQA / conformal prediction	irrelevant/noisy retrieval and uncertainty sets	coverage, set size, calibration	Good for uncertainty; less direct for free-form generation.
QueryRobust (Percin et al., 2025)	query robustness	QA with query perturbations	wording variation at query–retrieval interface	retrieval and end-task degradation	Focuses on query perturbation, not corpus corruption.
PoisonArena (Chen et al., 2025d)	adversarial poisoning	RAG attack competition setting	competing poisoning attacks	attack success, retrieval effects	Strong security focus; benign uncertainty is secondary.
SafeRAG (Liang et al., 2025b)	RAG security	security benchmark for RAG	poisoning / malicious retrieval contexts	attack and defense metrics	Security-oriented; not a full evidence reliability suite.
MTRAG-UN (Rosenthal et al., 2026)	multi-turn unanswerability	multi-turn RAG conversations	unanswerable, underspecified, non-standalone questions	abstention / clarification quality	Covers interaction; less focused on adversarial evidence.
CRUMQs (Liu et al., 2025a)	unanswerable multi-hop RAG	realistic multi-hop queries	unanswerable and uncheatable questions	answerability and robustness	Useful for abstention; not a broad attack/noise benchmark.

Table 12: Benchmark-level comparison for robust RAG evaluation. The table makes explicit what each benchmark perturbs, what it measures, and which failure modes remain outside its scope.

Benchmark component	Required annotation	Metric suite	Why it is needed
Evidence reliability	relevance, sufficiency, freshness, provenance, support span	retrieval recall, sufficient-context accuracy, support precision	Distinguishes merely related passages from answer-supporting evidence.
Distraction and conflict	distractor type, conflicting claim pair, source agreement	answer flip rate, conflict sensitivity, context-utilization score	Reveals whether models ignore harmful context or silently pick a source.
Adversarial corruption	attacker access, insertion budget, trigger type, poisoned document ID	attack success, poisoned retrieval rate, stealth, defense bypass	Makes security results comparable across threat models.
Unanswerability and ambiguity	answerability label, ambiguity type, clarification target	selective risk, abstention precision/recall, clarification accuracy	Evaluates robust action selection, not just answer generation.
Cost and system action	number of retrievals, verifier calls, latency, final action trace	cost-adjusted accuracy, action utility, coverage–risk curve	Prevents methods from hiding robustness gains behind unbounded computation.

Table 13: A proposed benchmark blueprint for robust RAG under unreliable evidence. This is a research agenda rather than a validated leaderboard; existing benchmarks cover parts of the schema.

Pilot item type	Evidence state label	Correct action trace	Answer metric alone would say	Action-aware diagnosis
Benign hard negative	Gold evidence present; hard negative ranked above it	retrieve → rerank/filter → answer with support	Correct/incorrect answer	Separates retriever ranking error from generator distraction.
Insufficient context	Retrieved context related but lacks answer-bearing span	retrieve → detect insufficiency → re-retrieve or abstain	Wrong answer or refusal	Rewards abstention/retrieval control rather than forced guessing.
Retrieved conflict	Two sources disagree with different dates/provenance	retrieve → compare sources → expose conflict or prefer fresher source	One answer may match gold	Checks whether the system hides unresolved conflict.
Poisoned evidence	Malicious document retrieved with plausible support	retrieve → detect/demote/quarantine → answer or abstain	Attack success/failure	Distinguishes poisoned retrieval, defense detection, and final answer steering.

Table 14: Illustrative action-trace pilot for the benchmark blueprint. This is a worked annotation example, not a released leaderboard or empirical system ranking.

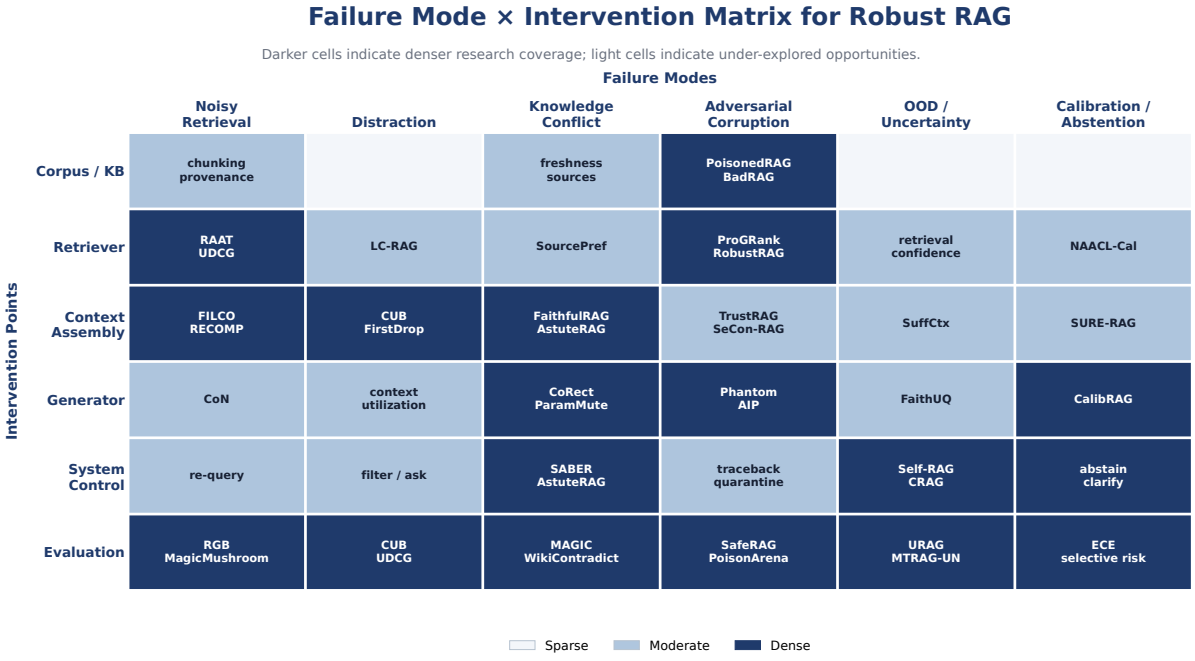


Figure 5: Failure mode × intervention matrix for robust RAG. Rows indicate where a method intervenes in the pipeline, while columns indicate the failure mode being addressed. Darker cells denote denser research coverage and lighter cells indicate under-explored opportunities.

Toy policy	Ans.	Act.	Joint
Answer-only	0.25	0.25	0.25
Abstain-on-missing	0.67	0.50	0.50
Evidence-action oracle	1.00	1.00	1.00

Table 15: Action-trace pilot over 12 synthetic diagnostic items. Answer accuracy, action accuracy, and joint answer-action accuracy diagnose different behavior.

Axis	Typical variation across papers	Why conclusions fail to transfer	Minimum reporting requirement
Retriever/generator setting	BM25 vs dense vs hybrid retrieval; small vs frontier generators; frozen vs fine-tuned models	A defense may work by compensating for one retriever or one model scale rather than solving the failure mode.	Retriever, index, top-k, reranker, generator, prompt, decoding, and context length.
Attack budget and access	Poisoned-document count, corpus write access, query control, retriever visibility, generator access	Attack success rates are incomparable when one attacker edits a corpus and another only controls the query.	Capability matrix, insertion budget, target set, stealth objective, and whether malicious evidence is retrieved.
Metric target	EM/F1, faithfulness, citation support, attack success, calibration error, selective risk, cost	Improving answer accuracy can worsen citation faithfulness; improving abstention can reduce coverage; reducing attacks can lower utility.	Report utility, evidence support, risk, calibration, cost, and security as a Pareto surface.
Statistical and reproducibility evidence	Single seed/model vs multiple seeds/models; code/data availability varies	Small prompt or model changes can reverse robustness rankings, especially for LLM-as-judge or attack benchmarks.	Seeds, confidence intervals or bootstrap tests where feasible, released prompts, data splits, and code or exact API settings.
Long-context baseline	RAG-only comparisons vs long-context LLMs with the same documents	A robustness gain may reflect retrieval compression rather than evidence reliability control.	Include a long-context or full-context baseline when context length permits, and report latency/cost.
Attribution vs answer accuracy	Some tasks score only final answer; others require evidence-supported citations	A model can answer correctly for the wrong reason, cite irrelevant evidence, or ignore retrieved support.	Claim-level support labels, citation precision/recall, and answer correctness reported separately.

Table 16: Evaluation-assumption matrix for robust RAG. The table summarizes why benchmark results often cannot be compared without aligning system setup, threat model, metric target, reproducibility evidence, and long-context baselines.

Backbone conclusion	Venue-linked evidence base	What remains stable	What remains preprint/trend evidence
Irrelevant or noisy evidence can change generation behavior.	Irrelevant-context robustness, noise-aware training, counterfactual-noise analyses, and robust fine-tuning papers in ICLR/ACL/NAACL/SIGIR-style venues.	Robustness is not just retriever recall; generator selectivity and context construction matter.	Fine-grained noise taxonomies, long-tail stress tests, and model-specific mitigation rankings.
Knowledge conflict is distinct from poisoning.	Conflict-resolution and imperfect-retrieval work in TACL/COLING/EMNLP-style records, plus venue-linked conflict benchmarks.	Natural contradiction requires source reconciliation and evidence sufficiency reasoning.	Many automatic conflict-resolution methods and large-scale conflict benchmark variants.
RAG security comparisons require capability normalization.	USENIX/WWW/CCS/KDD-style poisoning, traceback, membership-inference, and backdoor records.	Attack success cannot be interpreted without corpus write access, query control, retriever visibility, insertion budget, and stealth.	The detailed ranking of attacks/defenses and many newly proposed bypasses.
Calibration and abstention are action-control decisions.	Venue-linked relevance-assessment, answerability, calibration, and RAG evaluation records.	Confidence, evidence sufficiency, retrieve/answer decisions, and abstention are separable evaluation targets.	Specific adaptive-control policies and frontier-model calibration behavior.
Benchmark design needs evidence and action labels.	RGB, RAGTruth, RAG evaluation libraries, long-form/context benchmarks, and challenge-style records.	Answer accuracy alone is insufficient for robust RAG evaluation.	The full action-trace blueprint remains a proposal until implemented and validated.

Table 17: Peer-reviewed / venue-linked backbone of the survey. These are the conclusions the paper treats as stable when preprint-only evidence is downweighted.

Work	Failure	Task / setup	Key assumption	Metric	Boundary
IrrelCtx	irrelevant context	Open-domain QA	Irrelevance can be filtered or learned against	Robust accuracy	No adversarial/conflict setting.
RAAT	retrieval noise	Noisy QA training	Noise-aware training improves generation	F1/EM	Needs deployment-like noise.
RGB	multi-ability RAG	Bilingual benchmark	RAG reliability decomposes into abilities	Ability accuracy	Not adversarially adaptive.
FaithfulRAG	conflict	Fact-level conflict	Conflicts can be localized and reasoned over	Faithfulness	Needs reliable fact extraction.
AstuteRAG	imperfect retrieval	Training-free consolidation	Internal/external knowledge can be reliability-compared	Robust QA	Depends on LLM self-assessment.
PoisonedRAG	poisoning	Injected passages	Few crafted passages can dominate retrieval	Attack success	Targeted corruption focus.
RobustRAG	defense	Bounded corruption	Isolate-and-aggregate can certify robustness	Certificate, accuracy	Assumes attack budget.
Self-RAG / CtrlA	adaptive retrieval	Retrieve-or-answer	Model state can trigger retrieval	Trigger quality	Calibration can shift.
CalibRAG / URAG	calibration	Decision/MCQA	Confidence should reflect evidence uncertainty	ECE, coverage	Does not repair evidence.
PoisonArena / SafeRAG	security eval.	Poisoning benchmark	Threat models must be explicit	ASR, bypass	Still isolates attacks.

Table 18: Analytical matrix of representative robust RAG work. Compared with a citation list, the matrix highlights each work's task setup, central assumption, metric focus, and boundary of applicability.

Work	System setup	Failure construction	Metric / cost	Main claim	Failure boundary
IrrelCtx	RALM / open-domain QA; retrieved context injected into generator	Irrelevant passages with varying plausibility	Robust accuracy / answer degradation	Models can be brittle to irrelevant context.	Does not cover adversarial corruption or answerability policy.
RAAT	Noise-aware training for retrieval-augmented LMs	Multiple retrieval-noise types during training/evaluation	EM/F1 under noisy retrieval	Adaptive adversarial training improves noise robustness.	Transfer depends on whether deployment noise matches training noise.
RGB	Benchmark over English/Chinese QA-style RAG tasks	Noise, negative rejection, integration, counterfactual evidence	Ability-specific accuracy	RAG reliability decomposes into distinct abilities.	Benchmark abilities are diagnostic but not a unified control policy.
Sufficient Context	QA contexts labeled by sufficiency	Insufficient vs sufficient retrieved evidence	Selective answering / abstention quality	Evidence sufficiency should govern answerability.	Does not repair retrieval and does not model adversarial evidence.
PoisonedRAG	Corpus-injection attack against RAG	Crafted poisoned passages inserted into retrieval corpus	Attack success, retrieval of poison	Few poisoned documents can steer answers.	Strength depends on corpus write access, topic targeting, and retrieval success.
Certified / isolate defenses	Bounded-corruption RAG defense setting	Assumed corruption budget over retrieved evidence	Certificate, robust accuracy	Aggregation can certify robustness under budget assumptions.	Guarantees narrow when attacker violates budget or attacks instructions/tools.
Self-RAG / CtrlA	Adaptive retrieve-or-answer policies	Uncertainty about whether retrieval is needed	Trigger quality, answer utility, cost	Retrieval should be a conditional action.	Calibration shifts under distribution change; action cost often underreported.
RAGTruth	Annotated hallucination/faithfulness resource	Naturally generated unsupported spans across tasks	Span-level hallucination labels	Fine-grained support labels improve evaluation.	Faithfulness labels do not fully specify retrieval failure or system action.

Table 19: Method-level comparison of representative robust-RAG work. The table makes explicit the setup, failure construction, metric target, main claim, and failure boundary that must be aligned before comparing methods.

Case	Representative papers	Setup variables that must align	Metric conflict	Synthesis / failure boundary
Noise vs. distraction	IrrelCtx, RAAT, RGB, counterfactual-noise work, robust fine-tuning	Retriever type, top-k, distractor hardness, gold evidence position, generator scale, context length	Retrieval recall can improve while answer utility drops because more negatives enter context.	Treat retrieval quality and generator selectivity as different mechanisms; compare methods only when negative evidence type and rank position match.
Conflict vs. poisoning	Tug-of-War, FaithfulRAG, AstuteRAG, PoisonedRAG, one-shot poisoning, traceback work	Whether contradiction is natural or attacker-created; source freshness; corpus mutability; attacker budget; retriever visibility	Faithfulness/conflict metrics reward source reconciliation; attack metrics reward target steering or bypass.	Conflict methods should not be evaluated as poisoning defenses unless adversarial capabilities and malicious evidence retrieval are explicit.
Calibration vs. abstention	Sufficient Context, URAG, CalibRAG, Self-RAG, CtrlA, trust/abstain methods	Whether uncertainty is about answer correctness, evidence sufficiency, retrieval usefulness, or user-facing risk	ECE, selective risk, abstention precision, and retrieval-trigger quality can move in different directions.	Separate belief estimation from action policy; report coverage, cost, and final action alongside correctness.
Security defenses	Certified defenses, isolate-and-aggregate, poisoned-document detection, activation traceback, prompt-injection defenses	Corruption budget, corpus write access, black/white-box retriever, query control, generator access, detection time	Lower attack success can hide utility loss or unrealistic attack assumptions.	Defense claims transfer only within the stated capability envelope; prevention, detection, demotion, traceback, and recovery are different outcomes.

Table 20: Deeper case-study synthesis. Each case identifies variables that must be aligned before reported results or method rankings are comparable.

Apparent conclusion	What the framework changes	Non-trivial finding	Implication
"Method A is more robust to noise than method B."	Aligns noise type, rank position, gold-evidence availability, and intervention point.	Some methods solve generator distraction while others solve retriever defects; their scores are not competing estimates of the same property.	Report noise source and intervention point before ranking methods.
"Defense X lowers attack success."	Separates prevention, demotion, detection, traceback, and recovery.	A defense can improve traceback without lowering user-facing attack success, or certify bounded corruption without handling prompt-bearing documents.	Security papers should report outcome type, not only attack success rate.
"Calibration improves reliability."	Separates confidence in answer correctness from evidence sufficiency and action utility.	A calibrated answerer can still answer from insufficient evidence; an abstaining system can lower risk while reducing coverage.	Calibration metrics should be paired with evidence and action metrics.
"Benchmark Y is harder."	Compares evidence state, required action, cost budget, and metric target.	A benchmark can be hard because retrieval fails, because evidence conflicts, or because the correct action is refusal; these failures should not be pooled.	Leaderboards should expose diagnostic sub-scores instead of one robust-RAG score.

Table 21: Non-trivial findings produced by the comparability framework. These are not new empirical rankings; they are corrections to common interpretations of robust-RAG results.

Naive comparison	Why it is tempting	Normalization under our framework	Changed interpretation
IrrelCtx vs. RAAT vs. RGB as "noise robustness"	All report performance under noisy or irrelevant context.	Align negative type, rank position, whether gold evidence remains, intervention point, and metric.	IrrelCtx mainly tests generator selectivity, RAAT tests training against noise distributions, and RGB tests benchmark abilities; no single winner exists without fixing the evidence state.
Sufficient Context vs. Self-RAG as "abstention/retrieval control"	Both affect whether the system answers or retrieves.	Separate evidence-sufficiency prediction from learned retrieve/critique/action policy.	Sufficient Context evaluates whether evidence is enough; Self-RAG evaluates a policy for using retrieval. Their errors compose rather than substitute.
PoisonedRAG vs. certified RAG defenses	Both concern corrupted retrieved evidence.	Align attacker budget, corpus write access, retriever visibility, whether poison is retrieved, and whether output is steered.	A defense can be certified for bounded corruptions yet not address prompt-bearing malicious documents or unbounded corpus insertion.
RAGTruth vs. answer-accuracy benchmarks	Both evaluate generated outputs.	Separate final answer correctness from claim support and citation faithfulness.	A correct answer can still fail evidence reliability if it cites unsupported or irrelevant passages.
Long-context RAG benchmarks vs. retrieval-noise benchmarks	Both test evidence use under many tokens.	Align context length, evidence position, distractor density, and whether retrieval is actually required.	Long context is not a replacement for evidence reliability; it changes the failure mode from missing evidence to evidence selection and attention.
Calibration methods vs. abstention methods	Both reduce overconfident wrong answers.	Align confidence target: answer correctness, evidence sufficiency, retrieval usefulness, or action utility.	Better calibration need not imply better abstention, and abstention can improve risk while hurting coverage.

Table 22: Small re-analysis pilot showing how the framework changes interpretation of common robust-RAG comparisons. The goal is not to re-rank systems, but to identify which comparisons are invalid, partially valid, or complementary.

Main claim	Human-checked evidence basis	What the evidence supports	Boundary
Irrelevant or noisy context can harm generation.	Irrelevant-context robustness, RAAT, RGB, and counterfactual-noise papers were checked at method/evaluation level.	Noise affects both retrieval quality and generator context use.	Does not imply one robustness score covers all noise types.
Knowledge conflict differs from adversarial corruption.	Conflict papers and poisoning papers were checked for source of contradiction and evaluation target.	Natural disagreement and attacker-created contradiction require different assumptions.	Conflict-resolution results should not be read as poisoning-defense results.
RAG attacks require capability normalization.	Poisoning, backdoor, prompt-injection, traceback, and certification papers were checked for attacker access and metric target.	Attack/defense results are incomparable without corpus write access, query control, retriever visibility, and budget.	The survey does not rank attacks by real-world prevalence.
Calibration, abstention, and retrieval control are distinct.	Sufficient-context, uncertainty, adaptive-retrieval, and calibration papers were checked for decision target.	Confidence estimation, evidence sufficiency, retrieval triggering, and abstention optimize different quantities.	Does not validate a single universal control policy.
Action-trace evaluation is a benchmark agenda.	Existing benchmarks were checked for answer, evidence, action, cost, and threat fields.	Existing benchmarks cover pieces of the proposed schema.	No new leaderboard or empirical system ranking is claimed.

Table 23: Core-claim verification ledger. LLM-assisted citation audits were used only as internal triage; the claims above are stated at the level supported by human inspection of central cited papers.

Overlapping work	Typical interpretation in adjacent surveys	Reinterpretation under evidence reliability + action control	Different conclusion enabled
RGB / RAG evaluation benchmarks	A benchmark measuring multiple RAG abilities.	A diagnostic split between noisy evidence, negative rejection, integration, and counterfactual evidence states.	A single RAG score hides which system action failed: ignore, integrate, reject, or expose conflict.
Irrelevant-context robustness	A robustness method or context-filtering problem.	A generator-selectivity failure conditioned on retrieved but harmful context.	It should not be compared directly with retriever training unless the intervention point and context hardness are fixed.
Sufficient Context / answerability work	An abstention or uncertainty benchmark.	A control-policy problem: evidence sufficiency determines answer vs re-retrieve vs abstain.	Abstention accuracy is incomplete without evidence sufficiency and action-cost labels.
PoisonedRAG / RAG poisoning	A security attack surface.	An integrity failure whose strength depends on corpus write access, retrieval success, generator access, and insertion budget.	Attack success rates are not comparable until capability fields are normalized.
RAGTruth / hallucination evaluation	A faithfulness or hallucination annotation resource.	A separation between answer correctness, claim support, citation support, and evidence sufficiency.	Correct answers can still fail evidence reliability; citation faithfulness and answer accuracy should be reported separately.
Self-RAG / adaptive retrieval	A retrieval-augmented generation method.	A system-control policy that chooses retrieve, critique, answer, or continue under uncertainty.	The relevant evaluation target is not only answer quality but whether the policy chose the right action at the right cost.

Table 24: How the same papers are interpreted differently under the proposed framework. This table grounds the novelty claim in concrete reinterpretations rather than broad positioning statements.

Dimension	Distribution in the 268-paper final adjudicated evidence map
Publication year	2025: 139; 2026: 65; 2024: 54; 2023: 10
Source / venue type	arXiv/emerging preprint: 224; venue/proceedings-linked: 42; review-platform record: 1; other/uncategorized: 1
Failure categories	adversarial corruption: 132; knowledge conflict: 83; evaluation/benchmark: 77; noisy retrieval: 74; calibration/control: 70; distraction: 39; OOD/uncertainty: 25; multimodal/agentic/code: 16; survey-related: 6
Method family	defense/mitigation: 92; attack: 72; benchmark/evaluation: 53; control/calibration: 22; analysis: 17; survey background: 6; conflict modeling: 4; method: 2
Intervention point	corpus/KB: 66; generator: 62; evaluation: 37; context assembly: 35; mixed: 28; system control: 26; retriever/context: 13; not applicable: 1

Table 25: Descriptive corpus statistics for the final adjudicated evidence map. These are map statistics, not field-wide prevalence estimates.

Evidence stratum	Count
emerging preprint / arXiv-only	222
peer-reviewed / proceedings-linked	42
preprint with year tag	2
review-platform record	1
other / uncategorized	1

Table 26: Evidence-strength strata in the 268-paper final adjudicated evidence map. Venue-linked coverage improves after triage, but the map remains preprint-heavy.

	Noise	Distract	Conflict	Attack	OOD	Calib.	Eval.
Noise	74	27	27	10	3	26	19
Distract	27	39	11	3	0	5	11
Conflict	27	11	83	22	5	20	34
Attack	10	3	22	132	1	10	21
OOD	3	0	5	1	25	19	10
Calib.	26	5	20	10	19	70	22
Eval.	19	11	34	21	10	22	77

Table 27: Category overlap in the 268-paper final adjudicated evidence map. The overlaps show why single-label taxonomies are unstable, but the counts are map statistics rather than field prevalence estimates.

Method family	Noise	Distract	Conflict	Attack	OOD	Control	Eval.
attack	2	0	5	72	0	2	3
defense	42	22	37	40	2	19	9
benchmark	10	8	24	12	11	12	52
control	6	1	2	0	10	22	3
conflict	1	2	4	0	0	2	0
analysis	9	5	8	2	1	10	6
method	0	0	0	0	1	2	0

Table 28: Method-family by failure-type cross-tabulation over the 268-paper final adjudicated evidence map. The table describes the map, not field-wide prevalence.

Stage	**Count**	**Rule**
Initial retrieval	1,904 records	Keyword search over arXiv, ACL Anthology, OpenReview, USENIX, AAAI/ACM landing pages where open versions were available, plus curated robust/trustworthy RAG paper lists.
Open full-text corpus	1,445 PDFs	Downloaded only open PDFs; removed failed downloads, inaccessible files, and non-PDF artifacts.
Deduplicated candidates	1,399 papers	Deduplicated by normalized title and arXiv/proceedings identifiers; retained the most complete metadata record when preprint and proceedings versions overlapped.
Relevance screening	536 included papers	Included papers that study RAG robustness, reliability, uncertainty, calibration, faithfulness, attack/defense, or evaluation under unreliable evidence; excluded application-only RAG papers.
Core pool	112 papers	Papers directly shaping the taxonomy, threat/failure model, intervention mechanisms, or evaluation framework.
Core reading set	100 papers	Core papers used as main narrative anchors; the remaining 12 core-pool papers stay in the broader survey set.
Base survey set	300 papers	Core papers plus representative supporting work across all taxonomy branches before venue follow-up triage.
Venue-augmented scoping set	326 papers	Base survey set plus 26 venue-linked additions after follow-up triage.
Candidate evidence records	274 papers	Base evidence map of 250 papers plus 24 venue-linked additions, double-coded and adjudicated.
Final adjudicated evidence map	268 papers	Candidate evidence records retained after final adjudication; used for map statistics and trend analysis.

Table 29: Screening-flow summary for the current corpus snapshot. The supplementary material provides exact query strings, source-wise counts, deduplication rules, taxonomy codebook, per-paper metadata, exclusion log, and structured evidence map.

Set	N	Audit artifact	Selection rule and role
Retrieved records	1,904	`search_protocol.json`	Initial keyword and curated-list retrieval before PDF availability, deduplication, or relevance screening.
Open full-text PDFs	1,445	`pdfs/`	Records with open PDFs available for full-text inspection; inaccessible or non-PDF artifacts removed.
Deduplicated candidates	1,399	`all_reviewed_deduped.csv`	Normalized-title and identifier deduplication; one record retained for preprint/proceedings overlaps.
Relevant included papers	536	`all_reviewed_deduped.csv`	Papers with decision `core` (112) or `include` (424); background-adjacent papers (289) and excluded papers (574) are tracked separately.
Core pool	112	`decision=core` in `all_reviewed_deduped.csv`	Papers judged central to taxonomy, threat/failure models, intervention mechanisms, or evaluation design.
Core reading set	100	`core100_full.csv`	Main narrative anchors. The 12 core-pool papers not in this set remain in the base survey set rather than being discarded.
Base survey set	300	`survey300_full.csv`	Core pool plus representative supporting work across failure categories, method families, and evaluation settings before venue follow-up triage.
Venue-augmented scoping set	326	`survey_set_venue_audited.csv`	Base survey set plus 26 venue-linked papers added after follow-up triage.
Base evidence map	250	`evidence_map.csv`	Original structured map used before venue follow-up triage.
Candidate evidence records	274	round2 full adjudication CSV	Base map plus venue-linked additions, double-coded by humans and adjudicated.
Final adjudicated evidence map	268	`evidence_map_adjudicated.csv`	Candidate evidence records with final `include_in_evidence_map=1`; used for final map statistics.

Table 30: Corpus-set ledger. The table reconciles reported counts. Exact memberships are released as CSV files, including `adjudicated_evidence274_round2_final_full.csv`.

Artifact / claim source	Verification performed	What is coding- or metadata-assisted	How it is used in the paper
Main narrative claims and central citations	Author inspection of full paper text for papers used to support central claims, tables, and figures.	Initial prioritization and category suggestions were LLM-assisted.	Used for qualitative synthesis and case-study comparisons.
Core reading set	100 papers curated from the 112-paper core pool after full-text relevance checks for narrative fit.	Core-pool creation used relevance scores and pre-labels as triage signals.	Used as the main reading backbone; not claimed to be a reproducible top-100 ranking without author judgment.
268-paper final adjudicated evidence map	Metadata and structured fields manually cleaned; venue-candidate triage completed; 274 candidate records independently double-coded and adjudicated into 268 included evidence-map papers.	Initial candidate discovery and pre-labeling were LLM-assisted, but final evidence-map labels use human adjudication for coder disagreements.	Used for final map statistics and cross-tabs, with the caveat that the map is not a field census.
Human double-coding package	Two anonymized human sources independently coded all 274 candidate evidence records before discussion; round-2 agreement is strong across taxonomy, intervention, evidence-strength, relevance, core, and inclusion fields.	Disagreements were resolved in the final adjudication file.	Used to report pre-adjudication IAA and produce final adjudicated evidence-map labels.
Venue metadata pass and triage	ACL Anthology title/abstract scan plus DBLP/OpenAlex query logs, title matching, and follow-up triage decisions.	Discovery is metadata-based; inclusion/exclusion decisions are author triage rather than independent double coding.	Used to update venue-linked coverage and quantify remaining systematic-review limits.

Table 31: Verification and annotation provenance. The table separates full-text author inspection, coding-assisted labels, metadata-derived venue candidates, and non-adjudicated consistency diagnostics.

Item	What was done	What was hidden from annotators	What we do not claim
Round-2 coders	Two anonymized human annotation sources independently coded the same 274 candidate evidence records.	Machine pre-labels, prior evidence-map labels, previous adjudication logs, and the other source's labels.	We do not reveal personal identities in the submission; anonymized provenance is reported instead.
Training / codebook	Coders used the human double-coding protocol and codebook defining scope, failure type, method family, intervention point, evidence strength, and inclusion fields.	Prior LLM rationales and model confidence scores.	We do not claim the codebook is final for all RAG subareas; adjudication notes can still motivate revisions.
Independence	Round-2 IAA was computed before adjudication using independently completed CSV files.	Post-discussion labels were not used for IAA.	We do not use adjudicated labels to inflate agreement.
Adjudication	A final adjudication file resolved 469 field-level disagreements and produced a 268-paper evidence map.	Source labels were preserved and not overwritten.	We do not claim venue-complete coverage; adjudication only resolves labels for included candidate records.

Table 32: Annotation transparency for the human coding round. The table reports what can be audited from the blind annotation package without disclosing annotator identities.

Field	Agreement	κ / score	Interpretation
Full-text checked	100.0%	1.00	Both sources marked every evidence-map paper as full-text checked.
In-scope screening	99.6%	0.92	Near-complete agreement; only one screening disagreement remains for adjudication.
Relevance level	–	0.80	Linear-weighted agreement is strong for exclude/background/support/core priority.
Primary failure type	97.1%	0.96	Very strong human agreement on the main taxonomy category.
Method family	89.4%	0.86	Strong agreement on attack, benchmark, control, method, and analysis families.
Intervention point	93.4%	0.92	Very strong agreement on pipeline placement.
Evaluation type	94.9%	0.93	Very strong agreement on metric/evaluation family.
Evidence strength	97.8%	0.95	Very strong agreement on proceedings/preprint/review-platform strata.
Core-candidate decision	91.6%	0.80	Strong agreement on whether papers should shape the core narrative.
Include in evidence map	99.3%	0.85	Strong agreement on evidence-map inclusion.
Secondary labels	79.2% exact	0.83 Jaccard	Multilabel agreement is strong enough to support overlap analysis after adjudication.

Table 33: Round-2 human-human agreement on 274 candidate evidence records before adjudication. Two anonymized human annotation sources independently coded the same records; remaining disagreements were subsequently resolved to produce the 268-paper final adjudicated evidence map.

Claim	Status under venue-linked / peer-reviewed evidence	What changes without preprints	Interpretation
Retrieved evidence can harm generation even when relevant evidence exists.	Holds: supported by venue-linked work on irrelevant/noisy context, noise-robust training, counterfactual noise, and imperfect retrieval.	The range of stress-test variants shrinks.	This is a stable mechanism-level claim.
RAG security results require attacker-capability normalization.	Holds partially: venue-linked security work covers poisoning, traceback, membership inference, and backdoor-style corpus attacks.	Many fine-grained attack variants and defenses remain preprint-heavy.	The capability matrix is stable; detailed landscape counts are emerging trends.
Calibration, abstention, and retrieval control are distinct decisions.	Holds conceptually with venue-linked answerability, relevance-assessment, and calibration/control papers.	Specific method rankings and benchmark prevalence become sparse.	The conceptual separation remains; empirical prevalence is not settled.
The evidence map is attack- and benchmark-heavy.	Does not become a field-wide claim.	Under peer-reviewed-only evidence, the sample is much smaller and more evaluation-heavy.	Treat distributional cross-tabs as corpus diagnostics, not prevalence estimates.
Action-trace benchmark blueprint should score answer, evidence state, action, and cost.	Supported as a design synthesis, not validated as a benchmark standard.	The proposal becomes less empirical and more normative.	Keep as a research agenda unless a full benchmark is built and evaluated.

Table 34: Peer-reviewed / venue-linked sensitivity analysis. The table states which conclusions survive when preprint-only evidence is downweighted and which remain hypothesis-generating.

Pass	Scope	Observed yield	Interpretation
ACL Anthology metadata scan	ACL, EMNLP, NAACL, COLING; 2020–2026; full anthology BibTeX with abstracts	12,924 entries scanned; 181 RAG candidates; 30 robust-RAG candidates	A reproducible proceedings-level metadata scan for core NLP venues, not just backfilled metadata.
DBLP/OpenAlex metadata queries	15 robust-RAG query strings over venue-linked metadata for SIGIR, WWW, WSDM, KDD, USENIX Security, CCS, NDSS, ICLR, NeurIPS and overlaps	31 query rows; 51 unique venue-linked candidate titles after deduplication	Surfaced missing venue-linked candidates; all 51 were subsequently triaged.
Follow-up venue triage	51 venue-linked candidates	23 new evidence-map additions; 3 background additions; 9 exclusions; 2 duplicates; 13 already represented	Closes the unprocessed venue-candidate gap, while still not claiming complete manual proceedings coverage.
Final adjudicated map	268 evidence-map papers	42 venue/proceedings-linked; 222 arXiv-only or emerging preprints; 4 other strata	Improves peer-reviewed coverage but confirms that the map remains preprint-heavy.

Table 35: Venue-oriented search, coverage audit, and follow-up triage summary. Full query logs, candidate titles, decisions, and limitations are in `venue_systematic_queries.csv`, `venue_systematic_candidates.csv`, `venue_coverage_audit.csv`, and `venue_triage_decisions.csv`.

Coverage check	Count	Artifact	Interpretation
Venue-linked robust-RAG candidates found by metadata pass	51 unique titles	candidate CSV	Candidate pool from ACL Anthology, DBLP, and OpenAlex metadata queries.
Already in core100	6 / 51	coverage audit CSV	Core narrative captures only a small subset of venue-linked candidates.
Already in survey300	14 / 51	coverage audit CSV	The current survey set misses or intentionally excludes many venue-linked candidates.
Already in evidence map	13 / 51	coverage audit CSV	Cross-tabs should be interpreted as map statistics, not field statistics.
Not in original survey300	37 / 51	coverage audit CSV	These papers were subsequently triaged; see Table 38.

Table 36: Venue-coverage audit after the metadata pass. The audit quantifies residual coverage risk by matching venue-linked candidates against the original core, survey, and evidence-map sets. Files: `venue_systematic_candidates.csv` and `venue_coverage_audit.csv`.

Recall check	Observed result	What it controls	Residual risk
ACL Anthology metadata scan	12,924 entries scanned for ACL/EMNLP/NAACL/COLING; 181 RAG candidates; 30 robust-RAG candidates	Reduces reliance on arXiv-only discovery for core NLP venues.	Title/abstract terms can miss papers using older QA/IR terminology.
Cross-venue metadata pass	DBLP/OpenAlex queries yielded 51 venue-linked robust-RAG candidates	Adds IR, web, security, and ML venue-linked records.	API ranking, metadata naming, and transient query errors limit recall.
Follow-up triage	51 candidates triaged; 26 added to scoping set; 24 added to evidence map; 9 excluded; 2 duplicates	Closes the known unprocessed venue-candidate gap.	Does not prove complete coverage of every proceedings volume.
Evidence-strength sensitivity	Main conclusions separated into venue-linked backbone vs preprint trend evidence	Prevents preprint-heavy map statistics from becoming field-level claims.	Some emerging security/control subtopics remain too new for strong consensus.

Table 37: Venue-recall risk analysis. The venue pass improves auditability and adds missed venue-linked papers, but it is still not a manual venue-complete proceedings review.

Triage outcome	Count	Action	Reasoning
Already represented in base evidence map	13	retain	Candidate was already included; metadata pass only improved venue visibility.
New evidence-map additions	23	add to evidence extension	Directly studies robust RAG, RAG evaluation, evidence reliability, security, calibration, or robust generation.
Survey-set paper promoted to evidence extension	1	add to evidence extension	Already in the survey set but not the base map; venue audit identified it as relevant to trust/calibration.
Background additions	3	add to venue-augmented scoping set only	Relevant boundary papers on domain, multilingual, or multimodal RAG, but not central text evidence-reliability evidence.
Excluded false positives / applications	9	exclude with reason	Application or adjacent RAG papers without robustness/evidence-reliability analysis.
Duplicate venue records	2	deduplicate	Same work already represented through another metadata record.
Venue-augmented scoping set	**326**	base 300 + 26 additions	Updated scoping set after follow-up triage.
Final adjudicated evidence map	**268**	274 candidates - 6 exclusions	Final map after human adjudication of double-coded candidate evidence records.

Table 38: Follow-up triage of all 51 venue-linked candidates from the metadata pass. The 37 candidates outside the original survey set are no longer unprocessed; decisions and reasons are in `venue_triage_decisions.csv`.

Category	Definition	Inclusion rule
Noisy retrieval	Benign retrieval imperfection: irrelevant passages, hard negatives, missing positives, stale or low-quality evidence.	Explicitly analyzes or mitigates retrieval noise, not merely uses retrieval.
Distraction	Retrieved context has negative marginal utility and changes answer, confidence, or reasoning despite not supporting the answer.	Includes distractor, context-utilization, or long-context degradation analysis.
Knowledge conflict	Contradiction between parametric knowledge and retrieved evidence, or among retrieved sources.	Includes conflict detection, source preference, fact-level reconciliation, or contextual faithfulness.
Adversarial corruption	Adaptive attacker manipulates corpus, retriever, prompt template, retrieved context, memory, or tools.	Includes poisoning, prompt injection, backdoors, attack/defense benchmarks.
OOD / uncertainty	Query or evidence is ambiguous, underspecified, unanswerable, out-of-knowledge-base, or under distribution shift.	Includes answerability, uncertainty quantification, or selective prediction.
Calibration / control	System decides when to retrieve, trust evidence, answer, abstain, clarify, or escalate.	Includes adaptive retrieval, calibrated decision-making, or abstention.
Evaluation benchmark	Paper introduces benchmark, dataset, metric, or evaluation protocol for robust RAG.	Must define evaluation axis, data, perturbation/threat, or metric.

Table 39: Taxonomy codebook used for paper categorization. The full codebook is provided in the supplementary material.

Rank	Paper	Categories	Role
1	PoisonedRAG: Knowledge Corruption Attacks to Retrieval-Augmented Generation of Large Language Models	adv. corruption	attack
2	Certifiably Robust RAG against Retrieval Corruption	adv. corruption	defense
3	Making Retrieval-Augmented Language Models Robust to Irrelevant Context	noisy retrieval; distraction; calibration	method
4	Enhancing Noise Robustness of Retrieval-Augmented Language Models with Adaptive Adversarial Training	noisy retrieval; evaluation	defense
5	Phantom: General Backdoor Attacks on Retrieval Augmented Language Generation	adv. corruption	attack
6	URAG: A Benchmark for Uncertainty Quantification in Retrieval-Augmented Large Language Models	OOD/uncertainty; calibration; evaluation; noise	benchmark
7	Trust or Abstain? A Self-Aware RAG Approach	knowledge conflict; calibration; evaluation	method
8	The First Drop of Ink: Nonlinear Impact of Misleading Information in Long-Context Reasoning	distraction; noisy retrieval	analysis

Table 40: Sample entries from the core reading set. The full core set and metadata are provided in the supplementary material.

Paper	**Year**	**Failure / risk**	**Method family**	**Intervention**	**Boundary / note**
PoisonedRAG	2025	adversarial corruption	attack	corpus / KB	Targeted knowledge corruption; stealth and long-term dynamics vary.
RobustRAG	2024	adversarial corruption	defense	retriever / aggregation	Certification assumes bounded corruption.
IrrelCtx	2023	retrieval noise / distraction	defense	context filtering	Does not model adaptive attackers.
RAAT	2024	retrieval noise	training	generator	Requires deployment-like noisy training data.
FaithfulRAG	2025	knowledge conflict	conflict modeling	generator	Depends on fact extraction and alignment.
AstuteRAG	2025	imperfect retrieval + conflict	control / consolidation	system control	Relies on source-aware LLM self-assessment.
Self-RAG	2024	retrieval need	adaptive retrieval	system control	Retrieval trigger calibration can shift.
RGB	2024	multi-axis RAG failure	benchmark	evaluation	Not adversarially adaptive.
RAGTruth	2024	hallucination / faithfulness	benchmark	evaluation	Strong word-level labels, less retrieval-threat focus.
MTRAG-UN	2026	unanswerability	benchmark	system action	Multi-turn setting; security is secondary.

Table 41: Sample of the structured evidence map. The full CSV supplement contains title, authors, year, source, taxonomy categories, failure/risk type, method family, intervention point, benchmark/task, metrics, contribution, and limitation/note for the cited literature map.